Pure Modal Tableau

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Definition 0.0.1. (Similarity Type) A *similarity type* is a pair $\tau = (O, \rho)$ where O is a non-empty set of elements called modal terms. And where ρ is a function $O \to \mathbb{N}$ (called an arity function), which assigns each modal term to a natural number that denotes the number of arguments that the associated modal operator takes. We write α to denote a generic modal term.

Definition 0.0.2. (Completeness (1)) A (normal) logic Λ is called *strongly complete* with respect to a class of frames F if for every set of formulas Γ and every formula ϕ,

$$\Gamma \Vdash_{\mathsf{F}} \phi \text{ implies } \Gamma \vdash_{\Lambda} \phi.$$

Definition 0.0.3. (Completeness (2)) A (normal) logic Λ is *strongly complete* with respect to a class of frames F if for every set of formulas Γ:

$$\text{if } \Gamma \text{ is } \Lambda\text{-consistent then } \Gamma \text{ is satisfiable in some model based on a frame } \mathfrak{F} \in \mathsf{F}.$$

Proposition 0.0.4. *Completeness (1) (Definition 0.0.2) iff Completeness (2) (Definition 0.0.3)*

Chapter 1

Sound and Complete Tableau for Purely Modal languages

In this chapter, we consider modal languages that only use modal terms in which every connective, other than negation, is expressible in terms of modalities and negation, and all modalities have a uniform relational semantics parametric in a modal term. Moreover, modalities can be composed to form compound modalities, the relational semantics of which are given naturally and uniformly in terms of the composition of the accessibility relations corresponding to the modalities being composed. These languages are called pure modal languages. The language used in this thesis is adapted from [2] and is discussed in Section 1.1. All other sections report original research, unless stated otherwise. Section 1.2 introduces the tableau system **PMT** that we will be using, then we do a couple of examples in Section 1.3 to illustrate reasoning in purely modal languages. After this, we prove the soundness and completeness of **PMT** in Sections 1.4 and Section 1.5, respectively.

1.1 Introduction to Purely Modal Languages

Purely modal languages are modal languages that remove the use of the disjunction (and consequently the conjunction) and $\top$ (and consequently $\bot$). The pure modal language we investigate in this chapter is able to express everything in the basic modal language. We prove this in Proposition 1.1.12. For this, we define the similarity type that will be used (Definition 0.0.1) to be arbitrary, including the terms ι_0, ι_1, and ι_2, which are nullary, unary, and binary terms, respectively. For the language we are considering, we also allow modal terms to be composed under certain restrictions. This language is taken from [2]. Also, the definitions in this section are adapted from [2] as well. For our study, we restrict the language such that ι_0 is the only nullary modality we will consider. In this chapter, when we say τ is a similarity type, we mean that τ is a similarity type that includes the terms ι_0, ι_1, and ι_2.

Definition 1.1.1. (The set $MT(\tau)$) Let τ be a similarity type. We define the set $MT(\tau)$ (model terms from τ) inductively as follows:

label=(i) if $\alpha \in \tau$ then $\alpha \in MT(\tau)$,

lbbel=(ii) if $m > 0$ and $\alpha, \beta_1, \ldots, \beta_m \in MT(\tau)$ and $\rho(\alpha) = m$ then $\alpha(\beta_1, \ldots, \beta_m) \in MT(\tau)$ and $\rho(\alpha(\beta_1, \ldots, \beta_m)) = \rho(\beta_1) + \cdots + \rho(\beta_m)$,

lcbel=(iii) and nothing else.

Note: It is suitable to mention here the inductive definition of ι_n. We have the modal terms ι_0, ι_1, ι_2 in our similarity type, and so in $MT(\tau)$, we define $\iota_n = \iota_2(\iota_1, \iota_{n-1})$ ($n \geq 3$). We call the modal terms satisfying (i) atomic modal terms and those that satisfy (ii) composite modal terms.

Next, we define the frames we will be using. Note that if for any $(n+1)$-ary relations we will denote them as $R_\alpha(wu_1 \ldots u_n)$ or $R_\alpha wu_1 \ldots u_n$.

Definition 1.1.2. ($MT(\tau)$-frame) Let τ be a model similarity type, then a $MT(\tau)$-*frame* is a tuple $\mathfrak{F}_{MT(\tau)} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)})$ that consists of:

label=(i) A non-empty set W (of worlds).

lbbel=(ii) For each $m \geq 0$, each m-ary modal term α in $MT(\tau)$, an $(m+1)$-ary relation R_α, defined as follows:

- $R_{\iota_0} = W$, $R_{\iota_1} = \{(w, w) \mid w \in W\}$, $R_{\iota_2} = \{(w, w, w) \mid w \in W\}$,
- for every modal term $\alpha \in \tau$, we have $R_\alpha \subseteq W^{\rho(\alpha)+1}$,
- letting $\rho(\beta_i) = b_i, 1 \leq i \leq m$, $R_{\alpha(\beta_1,\ldots,\beta_m)} = \{(w, w_1^1, \ldots, w_{b_1}^1, w_1^2, \ldots, w_{b_2}^2, \ldots, w_1^m, \ldots, w_{b_m}^m) \in W^{b_1+b_2+\cdots+b_m+1} \mid$ there are $u_1, \ldots, u_m \in W$ where $R_\alpha w u_1 \ldots u_m$ and $R_{\beta_i} u_i w_1^i \ldots w_{b_i}^i$ for all $1 \leq i \leq m\} = R_\alpha \circ (R_{\beta_1}, \ldots, R_{\beta_m})$,
- and nothing else.

Note that when it is clear from the context, we often omit the subscript $MT(\tau)$ from $\mathfrak{F}_{MT(\tau)}$.

Definition 1.1.3. ($MT(\tau)$-model) Let $\mathfrak{F}_{MT(\tau)}$ be a $MT(\tau)$-frame. A $MT(\tau)$-*model* (denoted $\mathfrak{M}_{MT(\tau)}$) is an ordered pair $(\mathfrak{F}_{MT(\tau)}, V)$, where V : PVAR $\to \mathcal{P}(W)$ is a valuation on the set of propositional variables PVAR. We denote a $MT(\tau)$-model as $\mathfrak{M}_{MT(\tau)} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$, but again, we will often omit the subscript $MT(\tau)$ when it's clear from the context.

Now we introduce the purely modal language.

Definition 1.1.4. (Purely modal language) Let τ be a similarity type. The formulas in the purely modal language (which we denote as $\mathcal{L}_{P(\tau)}$) are given by the rule:

$$\varphi := p \mid \neg\varphi \mid [\alpha](\varphi_1, \ldots, \varphi_m)$$

where $p \in$ PVAR, $\alpha \in MT(\tau)$ and $\rho(\alpha) = m$.

Thus, the purely modal language is constructed from a set PVAR of propositional variables, negated formulas, and what we call boxed formulas.

Definition 1.1.5. (Dual of α) For any modal formula $[\alpha](\phi_1, \phi_2, \ldots, \phi_{\rho(\alpha)})$, we define its *dual* as $\langle\alpha\rangle(\phi_1, \phi_2, \ldots, \phi_{\rho(\alpha)}) := \neg[\alpha](\neg\phi_1, \neg\phi_2, \ldots, \neg\phi_{\rho(\alpha)})$.

Definition 1.1.6. (Semantics) Let $\mathfrak{M}_{MT(\tau)} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$ be a $MT(\tau)$-model and $w \in W$, then

- $\mathfrak{M}_{MT(\tau)}, w \Vdash p \quad$ iff $\quad p \in V(p)$
- $\mathfrak{M}_{MT(\tau)}, w \Vdash \neg\phi \quad$ iff $\quad \mathfrak{M}_{MT(\tau)}, w \nVdash \phi$

Now for the boxed formulas, we have two cases: $\rho(\alpha) = m > 0$ and $\rho(\alpha) = 0$.
So, if $\rho(\alpha) = m > 0$

$$\begin{gathered}\mathfrak{M}_{MT(\tau)}, w \Vdash [\alpha](\phi_1, \phi_2, \ldots, \phi_m) \\ \text{iff} \\ \text{for all } v_1, v_2, \ldots, v_m \in W \text{ with } R_\alpha w v_1 v_2 \ldots v_m \text{ we have} \\ \mathfrak{M}_{MT(\tau)}, v_1 \Vdash \phi_1 \text{ or } \mathfrak{M}_{MT(\tau)}, v_2 \Vdash \phi_2 \text{ or } \cdots \text{ or } \mathfrak{M}_{MT(\tau)}, v_m \Vdash \phi_m\end{gathered}$$

Now for ι_0, (i.e., when $m = 0$)

$$\mathfrak{M}_{MT(\tau)}, w \Vdash [\iota_0] \quad \text{iff} \quad w \in W$$

Remark 1. Now for the dual of the box ($m > 0$) using Definition 1.1.5:

$$\begin{gathered}\mathfrak{M}_{MT(\tau)}, w \Vdash \langle\alpha\rangle(\phi_1, \phi_2, \ldots, \phi_m) \\ \text{iff} \\ \text{there are states } v_1, v_2, \ldots, v_m \in W \text{ with } R_\alpha w v_1 v_2 \ldots v_m \text{ where} \\ \mathfrak{M}_{MT(\tau)}, v_i \Vdash \phi_i \text{ for each } i \in \{1, 2, \ldots, m\}\end{gathered}$$

and for $m = 0$:

$$\mathfrak{M}_{MT(\tau)}, w \Vdash \langle \iota_0 \rangle \quad \text{iff} \quad w \notin W$$

Note that from these definitions it follows that

$$[\alpha]([\beta_1](\phi_1^1, \phi_2^1, \ldots, \phi_{\rho(\beta_1)}^1), \ldots, [\beta_m](\phi_1^m, \ldots, \phi_{\rho(\beta_m)}^m))$$

is equivalent to

$$[\alpha(\beta_1, \ldots, \beta_{\rho(\alpha)})](\phi_1^1, \phi_2^1, \ldots, \phi_{\rho(\beta_1)}^1, \phi_1^2, \ldots, \phi_{\rho(\beta_2)}^2, \ldots, \phi_1^m, \ldots, \phi_{\rho(\beta_m)}^m).$$

Using the above equivalence, we can define a subset of formulas from the pure modal language, which we will call "Maximally composed" formulas. Where we compose as many boxes (or diamonds) as possible. We call this subset $\mathcal{L}_{P(\tau),max}$.

Definition 1.1.7. (Maximally composed formulas) The set of maximally composed $\mathcal{L}_{P(\tau)}$ formulas ($\mathcal{L}_{P(\tau),max}$) is defined by the union of the sets of formulas of the forms ϕ and ψ given by the mutually inductive rule:

$$\begin{aligned} \phi &:= p \mid \neg\psi \mid [\alpha](\psi_1, \psi_2, \ldots, \psi_{\rho(\alpha)}) \\ \psi &:= p \mid \neg\phi \mid \langle\alpha\rangle(\phi_1, \phi_2, \ldots, \phi_{\rho(\alpha)}) \end{aligned}$$

where $p \in \text{PVAR}$ and $\alpha \in MT(\tau)$.

The rest of this section describes how we can capture the semantics of the disjunction in $\mathcal{L}_{P(\tau)}$.

Capturing the semantics of the disjunction: Every similarity type τ contains the modal terms ι_0, ι_1, and ι_2 and so all $MT(\tau)$-models contain the respective corresponding relations. Using the definition that $\iota_n = \iota_2(\iota_1, \iota_{n-1})$ ($n \geq 3$), we can then conclude that the R_{ι_n} relation is $R_{\iota_n} = \{(\underbrace{w, \ldots, w}_{n+1\text{times}}) \mid w \in W\}$. So, for every $MT(\tau)$-model $\mathfrak{M}$:

$$\begin{gathered} \mathfrak{M}, w \Vdash [\alpha](\phi_1, \phi_2, \ldots, \phi_{\rho(\alpha)}) \\ \text{iff} \\ \text{for all } v_1, v_2, \ldots, v_{\rho(\alpha)} \text{ with } R_\alpha w v_1 v_2 \ldots v_{\rho(\alpha)} \text{ it is the case that} \\ \mathfrak{M}, v_1 \Vdash \phi_1 \text{ or } \mathfrak{M}, v_2 \Vdash \phi_2 \text{ or } \cdots \text{ or } \mathfrak{M}, v_{\rho(\alpha)} \Vdash \phi_{\rho(\alpha)}. \end{gathered}$$

This is for an arbitrary relation R_α, but if we use the ι-relation we get:

$$\begin{gathered} \mathfrak{M}, w \Vdash [\iota_n](\phi_1, \phi_2, \ldots, \phi_n) \\ \text{iff} \\ \text{for all } v_1, v_2, \ldots, v_n \text{ with } R_{\iota_n} w v_1 v_2 \ldots v_n \text{ it is the case that} \\ \mathfrak{M}, v_1 \Vdash \phi_1 \text{ or } \mathfrak{M}, v_2 \Vdash \phi_2 \text{ or } \cdots \text{ or } \mathfrak{M}, v_n \Vdash \phi_n. \end{gathered}$$

Now by definition, if $R_{\iota_n} w v_1 v_2 \ldots v_n$, then $(w, v_1, \ldots, v_n) \in R_{\iota_n}$. This means that $w = v_1 = \cdots = v_n$. So then:

$$\begin{gathered} \mathfrak{M}, w \Vdash [\iota_n](\phi_1, \phi_2, \ldots, \phi_n) \\ \text{iff} \\ \mathfrak{M}, w \Vdash \phi_1 \text{ or } \mathfrak{M}, w \Vdash \phi_2 \text{ or } \cdots \text{ or } \mathfrak{M}, w \Vdash \phi_n \\ \text{iff} \\ \mathfrak{M}, w \Vdash \phi_1 \vee \phi_2 \vee \cdots \vee \phi_n. \end{gathered}$$

As we can see, using the correct modality, we can capture the semantics of a disjunction. Now, note for the ι_1 and ι_0 we have that:

$$\begin{gathered} \mathfrak{M}, w \Vdash [\iota_1](\phi_1) \\ \text{iff} \\ \mathfrak{M}, w \Vdash \phi_1 \end{gathered}$$

and we have

$$\begin{array}{rll}\mathfrak{M}, w \Vdash [\iota_0] & \text{iff} & x \in R_{\iota_0} = W \\ & \text{iff} & \text{always} \\ & \text{iff} & \mathfrak{M}, w \Vdash \top.\end{array}$$

It may seem that ι_1 doesn't play much of a role, but it is useful since any formula ϕ is equivalent to $[\iota_1]\phi$. This equivalence will be used later. We also note that ι_0 captures the semantics of $\top$, which is used in the basic modal language.

Next, we want to prove that every modal language ML_τ is expressively equivalent to a purely modal language $\mathcal{L}_{P(\tau)}$. To do this, we will need to use induction on $\mathcal{L}_{P(\tau)}$, but due to how we can compose terms in $MT(\tau)$, it is helpful to introduce functions that measure the complexity of a formula. We construct such a function next.

Definition 1.1.8. (Cumulative Depth of a modal term) Let τ be a similarity type. We define the function $cep : MT(\tau) \to \mathbb{N}$ as follows:

1. if $\alpha \in MT(\tau)$ and α is atomic then $cep(\alpha) = 1$.
2. and if $\alpha \in MT(\tau)$ and α is composite, i.e., $\alpha = \pi(\beta_1, \dots, \beta_{\rho(\pi)}) \in MT(\tau)$ (for some $\pi, \beta_1, \dots, \beta_{\rho(\pi)} \in MT(\tau)$), then $cep(\alpha) = 1 + cep(\pi) + \Sigma_{i=1}^{\rho(\pi)} cep(\beta_i)$.

Definition 1.1.9. (Complexity of a formula) We let τ be a similarity type. We define the function $com : \mathcal{L}_{P(\tau)} \to \mathbb{N}$ as follows ($m = \rho(\alpha)$):

1. if $\phi = p$ then $com(\phi) = 0$
2. if $\phi = \neg\psi$ then $com(\phi) = 1 + com(\psi)$
3. if $\phi = [\alpha](\psi_1, \dots, \psi_m)$ then $com(\phi) = cep(\alpha) + \Sigma_{i=1}^m com(\psi_i)$.

Lemma 1.1.10. *Let $com : \mathcal{L}_{P(\tau)} \to \mathbb{N}$ be as defined above and let $\pi(\beta_1, \dots, \beta_m) \in MT(\tau)$ with $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$ for all $1 \le i \le m$. Then $com([\pi(\beta_1, \dots, \beta_m)](\phi_1^1, \dots, \phi_{b_1}^1, \dots, \phi_1^m, \dots, \phi_{b_m}^m))$ $> com([\pi]([\beta_1](\phi_1^1, \dots, \phi_{b_1}^1), \dots, [\beta_m](\phi_1^m, \dots, \phi_{b_m}^m)))$.*

Proof. Let $com : \mathcal{L}_{P(\tau)} \to \mathbb{N}$ be as defined above and let $\pi(\beta_1, \dots, \beta_m) \in MT(\tau)$ with $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$ for all $1 \le i \le m$. Now,

$$\begin{aligned}&com([\pi(\beta_1, \dots, \beta_m)](\phi_1^1, \dots, \phi_{b_1}^1, \dots, \phi_1^m, \dots, \phi_{b_m}^m)) \\ &\qquad = cep(\pi(\beta_1, \dots, \beta_m)) + \Sigma_{i=1}^m \Sigma_{j=1}^{b_i} com(\phi_j^i) \\ &\qquad = 1 + cep(\pi) + \Sigma_{i=1}^m cep(\beta_i) + \Sigma_{i=1}^m \Sigma_{j=1}^{b_i} com(\phi_j^i),\end{aligned}$$

and

$$\begin{aligned}&com([\pi]([\beta_1](\phi_1^1, \dots, \phi_{b_1}^1), \dots, [\beta_m](\phi_1^m, \dots, \phi_{b_m}^m))) \\ &\qquad = cep(\pi) + \Sigma_{i=1}^m com[\beta_i](\phi_1^i, \dots, \phi_{b_i}^i) \\ &\qquad = cep(\pi) + \Sigma_{i=1}^m (cep(\beta_i) + \Sigma_{j=1}^{b_i} com(\phi_j^i)) \\ &\qquad = cep(\pi) + \Sigma_{i=1}^m cep(\beta_i) + \Sigma_{i=1}^m \Sigma_{j=1}^{b_i} com(\phi_j^i).\end{aligned}$$

Hence, our original statement is satisfied. □

Lemma 1.1.11. *Every modality $\alpha \in MT(\tau)$ can be rewritten such that the leading modality is atomic.*

Proof. We do this by induction on $MT(\tau)$. *Base case:* If α is atomic, we are done. *Induction hypothesis:* Assume that the statement is true for any modality α such that $cep(\alpha) < m$. *Induction step:* Let α be composite such that $cep(\alpha) = m$. Now $\alpha = \pi(\beta_1, \dots, \beta_{\rho(\pi)})$ for some $\pi, \beta_1, \dots, \beta_{\rho(\pi)} \in MT(\tau)$. If π is atomic, we are done. If π is composite, since $cep(\pi) <$

$cep(\alpha)$, we can rewrite π as $\pi = \lambda(\mu_1, \ldots, \mu_{\rho(\lambda)})$ where λ is atomic using our induction hypothesis. Now $\alpha = (\lambda(\mu_1, \ldots, \mu_{\rho(\lambda)}))(\beta_1, \ldots, \beta_{\rho(\pi)})$, but we can reindex the β_i's (letting $\rho(\lambda) = l$ and $\rho(\mu_i) = m_i$ for all $1 \leq i \leq l$) to get $\alpha = (\lambda(\mu_1, \ldots, \mu_l))(\beta_1^1, \ldots, \beta_{m_1}^1, \ldots, \beta_1^l, \ldots, \beta_{m_l}^l) = \lambda(\mu_1(\beta_1^1, \ldots, \beta_{m_1}^1), \ldots, \mu_l(\beta_1^l, \ldots, \beta_{m_l}^l))$. Hence, by induction on $MT(\tau)$, every modality $\alpha \in MT(\tau)$ can be rewritten such that the leading modality is atomic. $\square$

Proposition 1.1.12. *Every modal language ML_τ is expressively equivalent to a purely modal language $\mathcal{L}_{P(\tau)}$.*

Proof. We begin by defining two translations $\zeta : ML_\tau \to \mathcal{L}_{P(\tau)}$ and $\eta : \mathcal{L}_{P(\tau)} \to ML_\tau$ by induction.

1. If $\phi = p \in \text{PVAR}$ then $\zeta(\phi) = p$
2. If $\phi = \top$ then $\zeta(\phi) = [\iota_0]$
3. If $\phi = \neg\psi$ then $\zeta(\phi) = \neg\zeta(\psi)$
4. If $\phi = \psi_1 \vee \psi_2$ then $\zeta(\phi) = [\iota_2](\zeta(\psi_1), \zeta(\psi_2))$
5. If $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ then $\zeta(\phi) = [\alpha](\zeta(\psi_1), \ldots, \zeta(\psi_{\rho(\alpha)}))$

and

1. If $\phi = p \in \text{PVAR}$ then $\eta(\phi) = p$
2. If $\phi = \neg\psi$ then $\eta(\phi) = \neg\eta(\psi)$
3. If $\phi = [\iota_0]$ then $\eta(\phi) = \top$
4. If $\phi = [\iota_1]\psi$ then $\eta(\phi) = \eta(\psi)$
5. If $\phi = [\iota_2](\psi_1, \psi_2)$ then $\eta(\phi) = \eta(\psi_1) \vee \eta(\psi_2)$
6. If $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ and α is atomic then $\eta(\phi) = [\alpha](\eta(\psi_1), \ldots, \eta(\psi_{\rho(\alpha)}))$
7. If $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ and α is composite then write α as $\pi(\beta_1, \ldots, \beta_{\rho(\pi)})$ where π is atomic (Lemma 1.1.11). After reindexing (letting $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$) $\phi = [\pi(\beta_1, \ldots, \beta_m)](\psi_1^1, \ldots, \psi_{b_1}^1, \ldots, \psi_1^m, \ldots, \psi_{b_m}^m)$ then $\eta(\phi) = [\pi](\eta([\beta_1](\psi_1^1, \ldots, \psi_{b_1}^1)), \ldots, \eta([\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m)))$.

Note that every $MT(\tau)$-model contains an ML_τ-model as a reduct, obtained by omitting the relations obtained by composition and only retaining those corresponding to terms in $\tau \setminus \{\iota_0, \iota_1, \iota_2\}$. We can therefore apply the definition of the semantics of ML_τ also to $MT(\tau)$ models. Now we are left to prove two statements:

1. That for every $MT(\tau)$-model $\mathfrak{M}$ and state w in $\mathfrak{M}$, and for every formula $\phi \in ML_\tau$, it is the case that $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$,
2. and that for every $MT(\tau)$-model $\mathfrak{M}$ and state w in $\mathfrak{M}$, and for every formula $\phi \in \mathcal{L}_{P(\tau)}$, it is the case that $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

We prove both of these by induction on ML_τ and $\mathcal{L}_{P(\tau)}$ respectively. So, for the first statement, let $\mathfrak{M} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$ be a $MT(\tau)$-model and w a state, and let $\phi \in ML_\tau$.

Case 1: If $\phi = p \in \text{PVAR}$ then $\mathfrak{M}, w \Vdash p$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$.

Case 2: If $\phi = \top$ then $\mathfrak{M}, w \Vdash \top$ iff $w \in W$ iff $\mathfrak{M}, w \Vdash [\iota_0]$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$.

Case 3: Assume the statement is true for ψ. If $\phi = \neg\psi$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash \neg\psi$ iff $\mathfrak{M}, w \nVdash \psi$ iff $\mathfrak{M}, w \nVdash \zeta(\psi)$ iff $\mathfrak{M}, w \Vdash \neg\zeta(\psi)$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$.

Case 4: Assume the statement is true for ψ_1 and ψ_2. If $\phi = \psi_1 \vee \psi_2$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash \psi_1 \vee \psi_2$ iff $\mathfrak{M}, w \Vdash \psi_1$ or $\mathfrak{M}, w \Vdash \psi_2$ iff $\mathfrak{M}, w \Vdash \zeta(\psi_1)$ or $\mathfrak{M}, w \Vdash \zeta(\psi_2)$ iff $\mathfrak{M}, w \Vdash [\iota_2](\zeta(\psi_1), \zeta(\psi_2))$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$.

Case 5: Let $\alpha \in \tau$ and assume the statement is true for $\psi_1, \ldots, \psi_{\rho(\alpha)}$. If $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ iff for all states $v_1, \ldots, v_{\rho(\alpha)}$ such that $R_\alpha w v_1 \ldots v_{\rho(\alpha)}$

and we have that for some $1 \leq i \leq \rho(\alpha)$, $\mathfrak{M}, v_i \Vdash \psi_i$. This is true iff $\mathfrak{M}, v_i \Vdash \zeta(\psi_i)$ for some $1 \leq i \leq \rho(\alpha)$ iff $\mathfrak{M}, w \Vdash [\alpha](\zeta(\psi_1), \ldots, \zeta(\psi_{\rho(\alpha)}))$ iff $\mathfrak{M}, w \Vdash \zeta(\phi)$.

For the second statement, let $\mathfrak{M} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$ be a $MT(\tau)$-model and w a state, and let $\phi \in \mathcal{L}_{P(\tau)}$.

Case 1: If $\phi = p \in$ PVAR then $\mathfrak{M}, w \Vdash p$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 2: Assume the statement is true for ψ. If $\phi = \neg\psi$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash \neg\psi$ iff $\mathfrak{M}, w \nVdash \psi$ iff $\mathfrak{M}, w \nVdash \eta(\psi)$ iff $\mathfrak{M}, w \Vdash \neg\eta(\psi)$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 3: If $\phi = [\iota_0]$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\iota_0]$ iff $w \in W$ iff $\mathfrak{M}, w \Vdash \top$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 4: Assume the statement is true for ψ. If $\phi = [\iota_1]\psi$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\iota_1]\psi$ iff $\mathfrak{M}, w \Vdash \psi$ (since $R_{\iota_1}ww$ and w is the only ι_1-successor) iff $\mathfrak{M}, w \Vdash \eta(\psi)$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 5: Assume the statement is true for ψ_1 and ψ_2. If $\phi = [\iota_2](\psi_1, \psi_2)$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\iota_2](\psi_1, \psi_2)$ iff $\mathfrak{M}, w \Vdash \psi_1$ or $\mathfrak{M}, w \Vdash \psi_2$ iff $\mathfrak{M}, w \Vdash \eta(\psi_1)$ or $\mathfrak{M}, w \Vdash \eta(\psi_2)$ iff $\mathfrak{M}, w \Vdash \eta(\psi_1) \vee \eta(\psi_2)$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 6: Let $\alpha \in MT(\tau)$ be atomic and assume the statement is true for $\psi_1, \ldots, \psi_{\rho(\alpha)}$. If $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ iff for all states $v_1, \ldots, v_{\rho(\alpha)}$ such that $R_\alpha w v_1 \ldots v_{\rho(\alpha)}$ and $\mathfrak{M}, v_i \Vdash \psi_i$, for some $1 \leq i \leq \rho(\alpha)$. This is true iff $\mathfrak{M}, v_i \Vdash \eta(\psi_i)$ for some $1 \leq i \leq \rho(\alpha)$, iff $\mathfrak{M}, w \Vdash [\alpha](\eta(\psi_1), \ldots, \eta(\psi_{\rho(\alpha)}))$ iff $\mathfrak{M}, w \Vdash \eta(\phi)$.

Case 7: Let $\alpha \in MT(\tau)$ be composite. Let $\phi = [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ and assume the statement is true for lower *com* value (Definition 1.1.9). Then $\mathfrak{M}, w \Vdash \phi$ iff $\mathfrak{M}, w \Vdash [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$. Now, since α is composite, we can rewrite it as a modal term $\pi(\beta_1, \ldots, \beta_{\rho(\pi)})$ where π is atomic (Lemma 1.1.11). After reindexing the ψ's (letting $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$) we get that $\mathfrak{M}, w \Vdash [\alpha](\psi_1, \ldots, \psi_{\rho(\alpha)})$ is true iff $\mathfrak{M}, w \Vdash [\pi(\beta_1, \ldots, \beta_m)](\psi_1^1, \ldots, \psi_{b_1}^1, \ldots, \psi_1^m, \ldots, \psi_{b_m}^m)$ iff

$$\mathfrak{M}, w \Vdash [\pi]([\beta_1](\psi_1^1, \ldots, \psi_{b_1}^1), \ldots, [\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m))$$

iff for any states $v_1, \ldots, v_m$ such that $R_\pi w v_1 \ldots v_m$ we have $\mathfrak{M}, v_i \Vdash [\beta_i](\psi_1^i, \ldots, \psi_{b_i}^i)$ for some $1 \leq i \leq m$. Since $com([\beta_i](\psi_1^i, \ldots, \psi_{b_i}^i)) < com(\phi)$, then by our induction hypothesis, this is true iff $\mathfrak{M}, v_i \Vdash \eta([\beta_i](\psi_1^i, \ldots, \psi_{b_i}^i))$ for some $1 \leq i \leq m$, iff

$$\mathfrak{M}, w \Vdash [\pi](\eta([\beta_1](\psi_1^1, \ldots, \psi_{b_1}^1)), \ldots, \eta([\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m)))$$

iff $\mathfrak{M}, w \Vdash \eta(\phi)$. □

Lemma 1.1.13. *Let $\mathfrak{M}$ and $\mathfrak{M}'$ be τ-models. Let $f : \mathfrak{M} \to \mathfrak{M}'$ be a mapping such that it satisfies the homomorphic condition (Definition* **??***) and the back condition (Definition* **??***) w.r.t. the relations corresponding to basic modal terms. Then f satisfies these conditions w.r.t. the relations corresponding to composed modal terms.*

Proof. Let $\mathfrak{M} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$ and $\mathfrak{M}' = (W', \{R'_\alpha\}_{\alpha \in MT(\tau)}, V')$ be τ-models. Let $f : \mathfrak{M} \to \mathfrak{M}'$ be such that for all atomic $\alpha \in MT(\tau)$, letting $\rho(\alpha) = m$, if $R_\alpha w_0 w_1 \ldots w_m$ then $R'_\alpha f(w_0) f(w_1) \ldots f(w_m)$, i.e., the homomorphic condition. Also let f be such that all atomic $\alpha \in MT(\tau)$ if $R'_\alpha f(w_0) w'_1 \ldots w'_n$ then there exist $w_1, \ldots, w_n$ such that $R_\alpha w w_1 \ldots w_n$ and $f(w_i) = w'_i$ (for $1 \leq i \leq n$), i.e., the back condition.

Now, we will prove our original statement by induction on $MT(\tau)$ — Definition 1.1.8. We already have our base case as an assumption. Now let $\alpha \in MT(\tau)$ be a composed modality. I.e., letting $\rho(\alpha) = n$, $\alpha = \pi(\beta_1, \ldots, \beta_{\rho(\pi)})$ and assume that $R_\alpha w_0 w_1 \ldots w_n$ and that the homomorphic condition is satisfied for any modality with *cep* value less than $cep(\alpha)$. Since $R_\alpha w_0 w_1 \ldots w_n$, we can rewrite this (letting $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$) as

$$R_{\pi(\beta_1, \ldots, \beta_m)} w_0 w_1^1 \ldots w_{b_1}^1 \ldots w_1^m \ldots w_{b_m}^m.$$

Therefore there are some states $v_1, \ldots, v_m$ such that $R_\pi w_0 v_1 \ldots v_m$ and $R_{\beta_i} v_i w_1^i \ldots w_{b_i}^i$ for all $1 \leq i \leq m$. Now since $cep(\pi) < cep(\alpha)$ and $cep(\beta_i) < cep(\alpha)$ for all $1 \leq i \leq m$, then by our induction hypothesis $R'_\pi f(w_0) f(v_1) \ldots f(v_m)$ and $R'_{\beta_i} f(v_i) f(w_1^i) \ldots f(w_{b_i}^i)$ for all $1 \leq i \leq m$. Hence

$$R'_{\pi(\beta_1, \ldots, \beta_m)} f(w_0) f(w_1^1) \ldots f(w_{b_1}^1) \ldots f(w_1^m) \ldots f(w_{b_m}^m).$$

Therefore $R'_\alpha f(w_0) f(w_1) \ldots f(w_n)$, satisfying the homomorphic condition.

Now, assume that the back condition is satisfied for any modality with *cep* value less than $cep(\alpha)$. Also, assume $R'_\alpha f(w_0)w'_1 \dots w'_n$. This we can reindex the w's (letting $\rho(\pi) = m$ and $\rho(\beta_i) = b_i$) as

$$R'_{\pi(\beta_1,\dots,\beta_m)} f(w_0) w'^1_1 \dots w'^1_{b_1} \dots w'^m_1 \dots w'^m_{b_m}.$$

Therefore there are some states $v'_1, \dots, v'_m$ such that $R'_\pi f(w_0)v'_1 \dots v'_m$ and $R'_{\beta_i} v'_i w'^i_1 \dots w'^i_{b_i}$ for all $1 \leq i \leq m$. We can see that $cep(\pi) < cep(\alpha)$ and so by our induction hypothesis we can say that there are states $v_1, \dots, v_m \in W$ such that

$$R_\pi w_0 v_1 \dots v_m \text{ and } f(v_i) = v'_i$$

for all $1 \leq i \leq m$. This means that $R'_{\beta_i} f(v_i) w'^i_1 \dots w'^i_{b_i}$ for all $1 \leq i \leq m$. Since $cep(\beta_i) < cep(\alpha)$ for all i's, we can apply the induction hypothesis on each i. Therefore, there are states $w^i_1, \dots, w^i_{b_i} \in W$ such that

$$R_{\beta_i} v_i w^i_1 \dots w^i_{b_i} \text{ and } f(w^i_j) = w'^i_j$$

for all $1 \leq i \leq m$, $1 \leq j \leq b_i$. Now, from $R_\pi w_0 v_1 \dots v_m$ and $R_{\beta_i} v_i w^i_1 \dots w^i_{b_i}$ for each $1 \leq i \leq m$, we can conclude that

$$R_{\pi(\beta_1,\dots,\beta_m)} w_0 w^1_1 \dots w^1_{b_1} \dots w^m_1 \dots w^m_{b_m}.$$

This shows that the back condition is satisfied since (after reindexing back) we have $R_\alpha w_0 w_1 \dots w_n$ and we have found $w_1, \dots, w_n$ such that $f(w_i) = w'_i$ for all $1 \leq i \leq n$. □

This is all we need for the pure modal language, and we will now introduce a tableau system for this language.

1.2 A tableau system for $\mathcal{L}_{P(\tau)}$

Sound and complete tableau systems give a useful way of testing modal formulae for their satisfiability or validity. They are used differently in different contexts, but in our context, they find models and states for which the given formulae (which are input into the tableau) will be true. Using this, we can also find if a modal formula is valid or not. The way we do this is: if you want to prove that ϕ is valid, input $\neg\phi$ into the tableau and if it cannot give you a model (i.e., the tableau closes) for $\neg\phi$, it means that there doesn't exist a model that satisfies $\neg\phi$, i.e., ϕ will be satisfied for any model and state, which means that it is valid. But how does a tableau do this? By systematically using inference rules, a formula is reduced to smaller parts until no inference rule can be applied. To make sure that the tableau works as needed, we need to prove that these rules are sound and complete, which we do in Sections 1.4 and 1.5. We will be using what is called a labelled tableau, where we assign states to the formulae in the tableau – an example will clarify what we mean by this. (See [4] for more on labelled tableau.) To continue from here, we need to define what a labelled tableau is. But there are a number of definitions that are needed before we can properly define this.

Definition 1.2.1. (Labelled formulae) First, we introduce a set 'Names' which is a possibly countably infinite set of special variables disjoint from PVAR that will be used as names for states in models. A *labelled* $\mathcal{L}_{P(\tau)}$ *formula* is an ordered pair (ϕ, x), denoted $\phi : x$ (read as ϕ at x) where $\phi \in \mathcal{L}_{P(\tau)}$ and $x \in$ Names.

Definition 1.2.2. (Relational atoms on τ) We define the set $\text{RNames}_\tau = \{R_\alpha \mid \alpha \in MT(\tau)\}$. A *relational atom on* τ is a $(\rho(\alpha)+2)$-tuple, $(R_\alpha, x_0, x_1, \dots, x_{\rho(\alpha)})$, denoted $R_\alpha(x_0 \dots x_{\rho(\alpha)})$ or $R_\alpha x_0 \dots x_{\rho(\alpha)}$, where $x_0, \dots, x_{\rho(\alpha)} \in$ Names (Definition 1.2.1) and $R_\alpha \in \text{RNames}_\tau$. We will often leave out the subscript when the context of what $MT(\tau)$ is clear.

Definition 1.2.3. (Language of tableau terms, $\mathcal{L}_{t(\tau)}$) We let $\mathcal{L}_{t(\tau)}$ be the set of all labelled $\mathcal{L}_{P(\tau)}$ formulae and all relational atoms on τ.

Definition 1.2.4. (Definition of Names(Δ)) Let $\Delta \subseteq \mathcal{L}_{t(\tau)}$. We define the set Names(Δ) as follows:

label=(i) If $\phi : x \in \Delta$ then $x \in \text{Names}(\Delta)$, lbbel=(ii) if $R_\alpha x_0 x_1 \dots x_n \in \Delta$ then $x_0, x_1, \dots, x_n \in \text{Names}(\Delta)$,

lcbel=(iii) and nothing else.

Definition 1.2.5. (Definition of RNames(Δ)) Let $\Delta \subseteq \mathcal{L}_{t(\tau)}$. We define the set RNames(Δ) as follows:

label=(i) if $R_\alpha x_0 x_1 \dots x_n \in \Delta$ then $R_\alpha \in \text{RNames}(\Delta)$,

lbbel=(ii) if $R_\gamma \in \text{RNames}(\Delta)$ and $\gamma = \alpha(\beta_1, \dots, \beta_n)$ then $R_\alpha, R_{\beta_1}, \dots, R_{\beta_n} \in \text{RNames}(\Delta)$,

lcbel=(iii) and nothing else.

Definition 1.2.6. (τ-theory) Let τ be a similarity type. We say a set of formulas from $\mathcal{L}_{P(\tau)}$ is called a *τ-theory*. I.e. Γ is a τ-theory if $\Gamma \subseteq \mathcal{L}_{P(\tau)}$.

The next two definitions are adapted from [3].

Definition 1.2.7. (Tableau rule) A *tableau rule* (β) consists of a premise $\mathcal{P}$ at the top and a finite number of conclusions $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_b$ below:

$$\begin{array}{c} \mathcal{P} \\ \mathcal{C}_1 \quad \mathcal{C}_2 \quad \cdots \quad \mathcal{C}_b \end{array} \quad (\beta)$$

The premise is a subset of $\mathcal{L}_{t(\tau)}$, and so is each conclusion.

Definition 1.2.8. (Labelled tableau) We define a *labelled tableau* for a set of labelled formulas Δ as a tree with the following conditions:

1. We set Δ as the root of the labelled tableau.
2. For any non-leaf (Δ') of the tree, its children are instantiations (i.e., when the schema is substituted for a concrete formula) of the conclusion of a tableau rule ($*$) where Δ' is an instantiation of the premise of ($*$)

Note: the nodes in these trees are sets. Also note that a *labelled tableau for a τ-theory* Γ is defined by setting the $\Delta = \{\phi : x \mid \phi \in \Gamma, x \in \text{Names}\}$.

Remark 2. This definition allows us to identify labelled tableau, but we construct tableaux dynamically by applying tableau rules to leaf nodes. A tableau rule with premise $\mathcal{P}$ can be applied to a node Δ' if Δ' is an instance of $\mathcal{P}$. We have the notion of an *end node*, which is a leaf node where we will not apply an inference rule. Also, if $\mathbf{b}$ is a branch of a tableau, then we denote $\mathbf{U}(\mathbf{b})$ as the union of the sets along path $\mathbf{b}$. Now, the steps to expand a tableau are as follows:

label=(i) Choose a leaf node Δ' and choose a rule (β) which is applicable to Δ'.

lbbel=(ii) If (β) has m conclusions then create m successor nodes for Δ', with each successor Δ_i carrying an appropriate instantiation of conclusion $\mathcal{C}_i$.

lcbel=(iii) We do this with the provision that if every rule application on a node Δ' (on a branch $\mathbf{b}$), will result at least one of its possible children Δ'', be such that $\Delta'' \subseteq \mathbf{U}(\mathbf{b})$, then Δ' is an end node, and so we don't expand the tableau. We also say that this branch (or node) is downward saturated (Definition 1.2.9)

ldbel=(iv) If a node Δ' is such that $\phi : x \in \Delta'$ and $\neg\phi : x \in \Delta'$, then that node is an end node, and we say that the *branch closes*. A branch also closes if $\neg[\iota_0] : x \in \Delta'$.

Now, note that in the definition of our tableau rules, any unaffected elements of $\mathcal{L}_{t(\tau)}$ will be copied down through the branch. This is known as Kracht's style. (For more on this, see [5]). But when displaying tableau, we will omit this copying of Δ as it is cumbersome. This is what we call Fitting's style [1]. Thus, when displaying these tableaux, we will only write the formula that the inference rule replaces/introduces.

Now if a branch meets condition d, we "close" the branch and we call it a *closed branch*. Any branch that doesn't close is called an *open branch*. A tableau closes if all its branches close. It is appropriate here to define what we mean by a branch and tableau being downward saturated.

Definition 1.2.9. (Downward saturated) Let $\mathbf{b}$ be a branch of a tableau and Δ any node in the branch. We call $\mathbf{b}$ *downward saturated* if for every rule $(*)$ (applicable to Δ) at least one of its children Δ' is such that $\Delta' \subseteq \mathbf{U}(\mathbf{b})$. We also say that a tableau is downward saturated if every branch is either closed or is downward saturated.

Now, we need to find a corresponding $MT(\tau)$-model using Names(Δ) and RNames(Δ). To do this, we define the mappings S and R. S associates each element in 'Names' to a state in W (in this model we are constructing). Let $\mathsf{S} : \text{Names}(\Delta) \to W$ be a mapping between these two sets. Where S takes a name (say x) and maps to some element in W. I.e, $\mathsf{S}(x) \in W$. (In most cases, we will use the names in the tableau to be what we call the states in W and only use this mapping formally when proving properties of our system.) R is defined as follows: $\mathsf{R} : \text{RNames}(\Delta) \to \{R^{\mathfrak{M}}_\alpha\}_{\alpha \in MT(\tau)}$, where $R_\alpha \overset{\mathsf{R}}{\longmapsto} R^{\mathfrak{M}}_\alpha$. Here $R^{\mathfrak{M}}_\alpha$ is the interpretation of R_α in the specific model $\mathfrak{M}$.

We now list out the rules for the tableau system we will be using. Note that Δ here denotes a subset of $\mathcal{L}_{t(\tau)}$ and $\rho(\alpha) = m$.

Negation Elimination:

$$\frac{\Delta\,, \neg\neg\phi : x}{\Delta\,, \phi : x} \qquad (\neg E)$$

Diamond Elimination:

$$\frac{\Delta\,, \neg[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x}{\Delta\,, R_\alpha x y_1 \ldots y_m\,, \neg\phi_1 : y_1\,, \ldots\,, \neg\phi_m : y_m} \qquad (\langle\alpha\rangle E)$$

Where $y_1, \ldots, y_m \in$ Names that are distinct from Names$(\Delta) \cup \{x\}$.

Box Elimination:

$$\frac{\Delta\,, [\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x\,, R_\alpha x y_1 \ldots y_m}{\Delta'\,, \phi_1 : y_1 \quad \cdots \quad \Delta'\,, \phi_m : y_m} \qquad ([\alpha]E)$$

Where Δ' stands for $\Delta\,, [\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x\,, R_\alpha x y_1 \ldots y_m$.

ι_n-relation on a state:

For some $0 \leq i, j \leq n \in \{1, 2\}, i \neq j$

$$\frac{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, \phi : x_i}{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, \phi : x_i\,, \phi : x_j} \qquad (R_{\iota_n} x)$$

ι_n-relation on any other relation:

For some $0 \leq i, j \leq n \in \{1, 2\}, i \neq j$

$$\frac{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, R_\alpha(b_0 b_1 \ldots x_i \ldots b_m)}{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, R_\alpha(b_0 b_1 \ldots x_i \ldots b_m)\,, R_\alpha(b_0 b_1 \ldots x_j \ldots b_m)} \qquad (R_{\iota_n} R_\alpha)$$

ι_1-relation introduction:

For some $x \in$ Names(Δ)

$$\frac{\Delta}{\Delta\,, R_{\iota_1}(xx)} \qquad (R_{\iota_1} I)$$

ι_2-relation introduction:

For some $x \in$ Names(Δ)

$$\frac{\Delta}{\Delta\,, R_{\iota_2}(xxx)} \qquad (R_{\iota_2} I)$$

Relation Reduction:

$$\frac{\Delta\,, R_{\alpha(\beta_1, \beta_2, \ldots, \beta_m)}(x x^1_1 x^1_2 \ldots x^1_{\rho(\beta_1)} x^2_1 \ldots x^2_{\rho(\beta_2)} \ldots x^m_1 \ldots x^m_{\rho(\beta_m)})}{\Delta\,, R_\alpha(x y_1 \ldots y_m)\,, R_{\beta_1}(y_1 x^1_1 \ldots x^1_{\rho(\beta_1)})\,, \ldots\,, R_{\beta_m}(y_m x^m_1 \ldots x^m_{\rho(\beta_m)})} \qquad (RR)$$

Where $y_1, \ldots, y_m \in$ Names that are distinct from
$\text{Names}(\Delta) \cup \{x, x_1^1, x_2^1, \ldots, x_{\rho(\beta_1)}^1, x_1^2, \ldots, x_{\rho(\beta_2)}^2, \ldots, x_1^m, \ldots, x_{\rho(\beta_m)}^m\}$
i.e., distinct from any names occurring in the premise.

Boxed Modality Distribution:

$$\frac{\Delta\,, [\alpha(\beta_1\,, \ldots\,, \beta_{\rho(\alpha)})](\phi_1^1\,, \phi_2^1\,, \ldots\,, \phi_{\rho(\beta_1)}^1\,, \phi_1^2\,, \ldots\,, \phi_{\rho(\beta_2)}^2\,, \ldots\,, \phi_1^m\,, \ldots\,, \phi_{\rho(\beta_m)}^m) : x}{\Delta\,, [\alpha]([\beta_1](\phi_1^1\,, \phi_2^1\,, \ldots\,, \phi_{\rho(\beta_1)}^1)\,, \ldots\,, [\beta_m](\phi_1^m\,, \ldots\,, \phi_{\rho(\beta_m)}^m)) : x} \quad ([\alpha(\beta)]D)$$

We will call this tableau system **PMT** (Pure-Modal-Tableau). We will prove the soundness and completeness of this system, but first, we need to define some notation before we can define the soundness and completeness of **PMT**.

Definition 1.2.10. (Δ satisfied in $\mathfrak{M}$) We say that $\Delta \subseteq \mathcal{L}_{t(\tau)}$ *is satisfied in* $\mathfrak{M} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$, if there is a mapping $\mathsf{S} : \text{Names}(\Delta) \to W$ and a mapping $\mathsf{R} : \text{RNames}(\Delta) \to \{R_\alpha\}_{\alpha \in MT(\tau)}$ such that:

label=(i) if $\phi : x \in \Delta$ then $\mathfrak{M}, \mathsf{S}(x) \Vdash \phi$,

lbbel=(ii) if $\mathrm{R}_\alpha x_0 x_1 \ldots x_n \in \Delta$ then $(\mathsf{S}(x_0), \mathsf{S}(x_1), \ldots, \mathsf{S}(x_n)) \in \mathsf{R}(R_\alpha) = R_\alpha^{\mathfrak{M}}$.

As a consequence of (ii) above, we also have that $\mathsf{R}(R_{\alpha(\beta_1,\ldots,\beta_n)}) = (\mathsf{R}(R_\alpha))\circ(\mathsf{R}(R_{\beta_1}), \ldots, \mathsf{R}(R_{\beta_n}))$. This is because $(\mathsf{R}(R_\alpha)) \circ (\mathsf{R}(R_{\beta_1}), \ldots, \mathsf{R}(R_{\beta_n})) = R_\alpha^{\mathfrak{M}} \circ (R_{\beta_1}^{\mathfrak{M}}, \ldots, R_{\beta_n}^{\mathfrak{M}}) = R_{\alpha(\beta_1,\ldots,\beta_n)}^{\mathfrak{M}}$.

Definition 1.2.11. (Δ is satisfiable) A set $\Delta \subseteq \mathcal{L}_{t(\tau)}$ is satisfiable if, there is a $MT(\tau)$-model $\mathfrak{M} = (W, \{R_\alpha\}_{\alpha \in MT(\tau)}, V)$, such that Δ is satisfied in $\mathfrak{M}$.

Definition 1.2.12. (Derivability in PMT) A formula ϕ is derivable from a finite set of formulas $\{\phi_1, \ldots, \phi_n\}$ if any tableau for $\{\phi_1 : x, \ldots, \phi_n : x, \neg\phi : x\}$ closes. This is denoted as:

$$\phi_1, \ldots, \phi_n \vdash_{\mathbf{PMT}} \phi$$

Now we extend this notion to a (possibly) infinite set Γ stipulating that:

$$\Gamma \vdash_{\mathbf{PMT}} \phi \text{ if for some finite subset } \{\phi_1, \ldots, \phi_n\} \subseteq \Gamma \text{ we have } \phi_1, \ldots, \phi_n \vdash_{\mathbf{PMT}} \phi.$$

We will prove the soundness and completeness of **PMT** in later sections, but for now, we apply these rules with a couple of examples. As mentioned earlier, we will not copy down formulas when displaying our tableau. Also note that for a relational atom $R_\alpha(x_1 x_2 \ldots x_{\rho(\alpha)})$, we often remove the brackets and write it as: $R_\alpha x_1 x_2 \ldots x_{\rho(\alpha)}$.

1.3 Examples using PMT

In this section we present several examples of derivations in **PMT**. We will assume the soundness (Theorem 1.4.1) and completeness (Theorem 1.5.5) of our system **PMT** for now, but this will be proven in subsequent sections. If the formula is not originally in $\mathcal{L}_{P(\tau)}$, we rewrite it so that it is $\mathcal{L}_{P(\tau)}$ — Proposition 1.1.12. Generally, when we input a formula into a tableau, we first compose any composable modalities so that it is in $\mathcal{L}_{P(\tau),max}$ as per Definition 1.1.7. This is not required, and we will do an example where we don't compose all modalities.

The first example we will do is an equivalent form of the K-axiom in $\mathcal{L}_{P(\tau)}$. We would expect the tableau to show that the K-axiom is valid. When we input the negation of a valid formula into a tableau, assuming soundness, the tableau will close.

Example 1.3.1. We start by rewriting the K-axiom as a formula in $\mathcal{L}_{P(\tau)}$ and then composing any modalities.

$$
\begin{aligned}
[\mathbf{1}](p \to q) \to ([\mathbf{1}]p \to [\mathbf{1}]q) &\equiv [\mathbf{1}](\neg p \vee q) \to (\neg[\mathbf{1}]p \vee [\mathbf{1}]q) \\
&\equiv \neg[\mathbf{1}](\neg p \vee q) \vee (\neg[\mathbf{1}]p \vee [\mathbf{1}]q) \\
&\equiv \neg[\mathbf{1}]([\iota_2](\neg p, q)) \vee [\iota_2](\neg[\mathbf{1}]p, [\mathbf{1}]q) \\
&\equiv \neg[\mathbf{1}(\iota_2)](\neg p, q) \vee [\iota_2]([\iota_1](\neg[\mathbf{1}]p), [\mathbf{1}]q) \\
&\equiv \neg[\mathbf{1}(\iota_2)](\neg p, q) \vee [\iota_2(\iota_1, \mathbf{1})](\neg[\mathbf{1}]p, q) \\
&\equiv [\iota_2](\neg[\mathbf{1}(\iota_2)](\neg p, q), [\iota_2(\iota_1, \mathbf{1})](\neg[\mathbf{1}]p, q)) \\
&\equiv [\iota_2]([\iota_1](\neg[\mathbf{1}(\iota_2)](\neg p, q)), [\iota_2(\iota_1, \mathbf{1})](\neg[\mathbf{1}]p, q)) \\
&\equiv [\iota_2(\iota_1, \iota_2(\iota_1, \mathbf{1}))](\neg[\mathbf{1}(\iota_2)](\neg p, q), \neg[\mathbf{1}]p, q)
\end{aligned}
$$

Now, let's input the negation of this and prove that it is valid:

(1)	$\neg[\iota_2(\iota_1, \iota_2(\iota_1, \mathbf{1}))](\neg[\mathbf{1}(\iota_2)](\neg p, q), \neg[\mathbf{1}]p, q) : x$	
(2)	$R_{\iota_2(\iota_1, \iota_2(\iota_1, \mathbf{1}))}xy_1y_2y_3, \neg\neg[\mathbf{1}(\iota_2)](\neg p, q) : y_1, \neg\neg[\mathbf{1}]p : y_2, \neg q : y_3$	$(\langle\alpha\rangle E)$ from (1)
(3)	$[\mathbf{1}(\iota_2)](\neg p, q) : y_1$	$(\neg E)$ from (2)
(4)	$[\mathbf{1}]([\iota_2](\neg p, q)) : y_1$	$([\alpha(\beta)]D)$ from (3)
(5)	$[\mathbf{1}]p : y_2$	$(\neg E)$ from (2)
(6)	$R_{\iota_2}xz_1z_2, R_{\iota_1}z_1y_1, R_{\iota_2(\iota_1,\mathbf{1})}z_2y_2y_3$	(RR) from (2)
(7)	$R_{\iota_2}z_2w_1w_2, R_{\iota_1}w_1y_2, R_{\mathbf{1}}w_2y_3$	(RR) from (6)
(8)	$[\mathbf{1}]p : w_1$	$(R_{\iota_n}x)$ from (7) and (5)
(9)	$[\mathbf{1}]p : w_2$	$(R_{\iota_n}x)$ from (7) and (8)
(10)	$p : y_3$	$([\alpha]E)$ from (7) and (9)
(11)	$[\mathbf{1}]([\iota_2](\neg p, q)) : z_1$	$(R_{\iota_n}x)$ from (6) and (4)
(12)	$[\mathbf{1}]([\iota_2](\neg p, q)) : z_2$	$(R_{\iota_n}x)$ from (6) and (11)
(13)	$[\mathbf{1}]([\iota_2](\neg p, q)) : w_2$	$(R_{\iota_n}x)$ from (7) and (12)
(14)	$[\iota_2](\neg p, q) : y_3$	$([\alpha]E)$ from (7) and (13)
(15)	$R_{\iota_2}y_3y_3y_3$	$(R_{\iota_2}I)$
(16)	$\begin{array}{cc} \neg p : y_3 & q : y_3 \\ \otimes & \otimes \end{array}$	$([\alpha]E)$ from (15) and (14)

Here (16a) closes w.r.t. (10) and (16b) closes w.r.t. (2). Therefore, the tableau is closed, i.e. it could not find a counter model for the K-axiom. Hence, we can conclude that the K-axiom is valid (using soundness, Theorem 1.4.1).

Example 1.3.2. In this example, we rewrite $\Diamond\top \to (\Box p \to \Diamond p)$ as a formula in $\mathcal{L}_{P(\tau),max}$, and then produce a **PMT** tableau deriving it.

$$
\begin{aligned}
\neg(\langle\mathbf{1}\rangle[\iota_0] \to ([\mathbf{1}]p \to \langle\mathbf{1}\rangle(p))) &\equiv \neg(\neg\langle\mathbf{1}\rangle[\iota_0] \vee (\neg[\mathbf{1}]p \vee \langle\mathbf{1}\rangle p)) \\
&\equiv \neg(\neg\neg[\mathbf{1}]\neg[\iota_0] \vee (\neg[\mathbf{1}]p \vee \neg[\mathbf{1}]\neg p)) \\
&\equiv \neg([\mathbf{1}]\neg[\iota_0] \vee (\neg[\mathbf{1}]p \vee \neg[\mathbf{1}]\neg p)) \\
&\equiv \neg([\mathbf{1}]\neg[\iota_0] \vee \neg[\mathbf{1}]p \vee \neg[\mathbf{1}]\neg p) \\
&\equiv \neg([\iota_3]([\mathbf{1}]\neg[\iota_0], \neg[\mathbf{1}]p, \neg[\mathbf{1}]\neg p)) \\
&\equiv \neg([\iota_3]([\mathbf{1}]\neg[\iota_0], [\iota_1](\neg[\mathbf{1}]p), [\iota_1](\neg[\mathbf{1}]\neg p))) \\
&\equiv \neg([\iota_3(\mathbf{1}, \iota_1, \iota_1)](\neg[\iota_0], \neg[\mathbf{1}]p, \neg[\mathbf{1}]\neg p)) \\
&\equiv \neg[\iota_3(\mathbf{1}, \iota_1, \iota_1)](\neg[\iota_0], \neg[\mathbf{1}]p, \neg[\mathbf{1}]\neg p)
\end{aligned}
$$

Although the formula has now been written to be in $\mathcal{L}_{P(\tau),max}$, **PMT** does not handle ι_n for any $n > 2$, so we need to reduce these with ι_1's and ι_2's.

$$\begin{aligned}\neg[\iota_3(\mathbf{1},\iota_1,\iota_1)](\neg[\iota_0],\neg[\mathbf{1}]p,\neg[\mathbf{1}]\neg p) &\equiv \neg[\iota_2(\iota_1(\mathbf{1}),\iota_2(\iota_1,\iota_1))](\neg[\iota_0],\neg[\mathbf{1}]p,\neg[\mathbf{1}](\neg p))\\ &\equiv \neg[\iota_2(\mathbf{1},\iota_2))](\neg[\iota_0],\neg[\mathbf{1}]p,\neg[\mathbf{1}]\neg p)\end{aligned}$$

The last step is convenient, but not required.

(1)	$\neg[\iota_2(\mathbf{1}\,,\iota_2)](\neg[\iota_0]\,,\neg[\mathbf{1}]p\,,\neg[\mathbf{1}]\neg p) : x$	
(2)	$R_{\iota_2(\mathbf{1},\iota_2)}xy_1y_2y_3\,,\neg\neg[\iota_0] : y_1\,,\neg\neg[\mathbf{1}]p : y_2\,,\neg\neg[\mathbf{1}]\neg p : y_3$	$(\langle\alpha\rangle E)$ from (1)
(3)	$[\mathbf{1}]p : y_2$	$(\neg E)$ from (2)
(4)	$[\mathbf{1}]\neg p : y_3$	$(\neg E)$ from (2)
(5)	$R_{\iota_2}xz_1z_2\,,R_{\mathbf{1}}z_1y_1\,,R_{\iota_2}z_2y_2y_3$	(RR) from (2)
(6)	$R_{\mathbf{1}}xy_1$	$(R_{\iota_n}R_\alpha)$ from (5)
(7)	$[\mathbf{1}]p : z_2$	$(R_{\iota_n}x)$ from (5) and (3)
(8)	$[\mathbf{1}]\neg p : z_2$	$(R_{\iota_n}x)$ from (5) and (4)
(9)	$[\mathbf{1}]p : x$	$(R_{\iota_n}x)$ from (5) and (7)
(10)	$[\mathbf{1}]\neg p : x$	$(R_{\iota_n}x)$ from (5) and (8)
(11)	$\neg p : y_1$	$([\alpha]E)$ from (6) and (10)
(12)	$p : y_1$	$([\alpha]E)$ from (6) and (9)
	$\otimes$	

Let's test the validity of a formula that does not belong in the basic modal language.

Example 1.3.3. For this example, we prove the validity of $[\mathbf{2}](p,q)\vee\langle\mathbf{2}\rangle(\neg p,\neg q)$, but we won't be composing every possible modality.

$$\begin{aligned}\neg([\mathbf{2}](p,q)\vee\langle\mathbf{2}\rangle(\neg p,\neg q)) &\equiv \neg([\iota_2]([\mathbf{2}](p,q),\langle\mathbf{2}\rangle(\neg p,\neg q)))\\ &\equiv \neg([\iota_2]([\mathbf{2}](p,q),\neg[\mathbf{2}](\neg\neg p,\neg\neg q)))\\ &\equiv \neg[\iota_2]([\mathbf{2}](p,q),\neg[\mathbf{2}](p,q))\end{aligned}$$

(1)	$\neg[\iota_2]([\mathbf{2}](p\,,q)\,,\neg[\mathbf{2}](p\,,q)) : x$	
(2)	$R_{\iota_2}xy_1y_2\,,\neg[\mathbf{2}](p\,,q) : y_1\,,\neg\neg[\mathbf{2}](p\,,q) : y_2$	$(\langle\alpha\rangle E)$ from (1)
(3)	$[\mathbf{2}](p\,,q) : y_2$	$(\neg E)$ from (2)
(4)	$[\mathbf{2}](p\,,q) : x$	$(R_{\iota_n}x)$ from (2) and (3)
(5)	$\neg[\mathbf{2}](p\,,q) : x$	$(R_{\iota_n}x)$ from (2)
	$\otimes$	

The node (5) closes with respect to (4). Since the tableau could not find a counter-model for $[\mathbf{2}](p,q)\vee\langle\mathbf{2}\rangle(\neg p,\neg q)$, then, with soundness, it is valid. Note that $[\mathbf{2}](p,q)\vee\langle\mathbf{2}\rangle(\neg p,\neg q)$ can be rewritten as $[\mathbf{2}](p,q)\vee\neg[\mathbf{2}](p,q)$, showing that $[\mathbf{2}](p,q)\vee\langle\mathbf{2}\rangle(\neg p,\neg q)$ follows the same scheme as the propositional formula $p\vee\neg p$. This is not a pure modal language instance of $p\vee\neg p$ since $\vee$ is not in $\mathcal{L}_{P(\tau)}$.

We can use this example to show that the tableau for a formula is not unique. Since there is no priority of rules that we need to use, we could have implemented the $(\langle\alpha\rangle E)$ rule to $\neg[\mathbf{2}](p,q) : y_1$, and the tableau would have taken more steps to close.

Remark 3. It is appropriate here to mention that the closure of tableau is independent of the order of rule application.

Before we go into the last example of this section, we will point out a more systematic way of keeping track of what formulas and relational atoms are in our set Δ. Since we don't copy all the previous conclusions over (when displaying tableaux), a more systematic way of doing this is necessary — especially when the tableau gets long.

For every tableau rule, there are labelled formulas and relational atoms that get replaced, and others get copied. When doing a tableau (using Fitting's style [1] to display these tableaux), we can tick off any labelled formulas/relational atoms that were replaced by the tableau rule. I.e., anything copied by the tableau rule is not ticked off. Let's see how this works when applying one rule.

If we have the following at a node:

$$\Delta\,, \neg[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x$$

Then applying the $(\langle\alpha\rangle E)$ rule, $\neg[\alpha](\phi_1, \phi_2\,, \ldots\,, \phi_m)$ is replaced by $R_\alpha x y_1 \ldots y_m\,, \neg\phi_1 : y_1\,, \ldots\,,$ $\neg\phi_m : y_m$ and so we can tick off $\neg[\alpha](\phi_1, \phi_2\,, \ldots\,, \phi_m)$. And in the tableau, it will look as follows:

$$\begin{array}{cc} \Delta\,, \neg[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x \checkmark & \\ \Delta\,, R_\alpha x y_1 \ldots y_m\,, \neg\phi_1 : y_1\,, \ldots\,, \neg\phi_m : y_m & (\langle\alpha\rangle E) \end{array}$$

Note: we have been doing this implicitly in the previous examples, so this is not describing a new way of doing these tableaux. So, let's do an example with this in mind:

Example 1.3.4. With this example we will input the negation of $\neg(\langle\mathbf{1}\rangle(p)\wedge(\langle\mathbf{1}\rangle(q)\wedge[\mathbf{1}](\neg(p\wedge q))))$ to deduce if this formula is valid. Again, we assume the soundness and completeness of **PMT**.

$$\begin{aligned} \neg\neg(\langle\mathbf{1}\rangle(p) \wedge (\langle\mathbf{1}\rangle(q) \wedge [\mathbf{1}](\neg(p \wedge q)))) &\equiv \langle\mathbf{1}\rangle(p) \wedge (\langle\mathbf{1}\rangle(q) \wedge [\mathbf{1}](\neg(p \wedge q))) \\ &\equiv \langle\mathbf{1}\rangle(p) \wedge (\langle\mathbf{1}\rangle(q) \wedge [\mathbf{1}](\neg p \vee \neg q)) \\ &\equiv \langle\mathbf{1}\rangle(p) \wedge (\langle\mathbf{1}\rangle(q) \wedge [\mathbf{1}]([\iota_2](\neg p\,, \neg q))) \\ &\equiv \langle\mathbf{1}\rangle(p) \wedge (\langle\mathbf{1}\rangle(q) \wedge [\mathbf{1}(\iota_2)](\neg p, \neg q)) \\ &\equiv \langle\mathbf{1}\rangle(p) \wedge \langle\iota_2\rangle(\langle\mathbf{1}\rangle(q), [\mathbf{1}(\iota_2)](\neg p, \neg q)) \\ &\equiv \langle\iota_2\rangle(\langle\mathbf{1}\rangle(p), \langle\iota_2\rangle(\langle\mathbf{1}\rangle(q), [\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \langle\iota_2\rangle(\langle\mathbf{1}\rangle(p), \langle\iota_2\rangle(\neg[\mathbf{1}](\neg q), [\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \langle\iota_2\rangle(\langle\mathbf{1}\rangle(p), \neg[\iota_2](\neg\neg[\mathbf{1}](\neg q), \neg[\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \langle\iota_2\rangle(\neg[\mathbf{1}](\neg p), \neg[\iota_2]([\mathbf{1}](\neg q), \neg[\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \neg[\iota_2](\neg\neg[\mathbf{1}](\neg p), \neg\neg[\iota_2]([\mathbf{1}](\neg q), \neg[\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \neg[\iota_2]([\mathbf{1}](\neg p), [\iota_2]([\mathbf{1}](\neg q), \neg[\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \neg[\iota_2]([\mathbf{1}](\neg p), [\iota_2]([\mathbf{1}](\neg q), [\iota_1](\neg[\mathbf{1}(\iota_2)](\neg p, \neg q)))) \\ &\equiv \neg[\iota_2]([\mathbf{1}](\neg p), [\iota_2(\mathbf{1}, \iota_1)](\neg q, \neg[\mathbf{1}(\iota_2)](\neg p, \neg q))) \\ &\equiv \neg[\iota_2(\mathbf{1}, \iota_2(\mathbf{1}, \iota_1))](\neg p, \neg q, \neg[\mathbf{1}(\iota_2)](\neg p, \neg q)) \end{aligned}$$

(1)	$\neg[\iota_2(\mathbf{1}\,,\iota_2(\mathbf{1}\,,\iota_1))](\neg p\,,\neg q\,,\neg[\mathbf{1}(\iota_2)](\neg p\,,\neg q)) : x\checkmark$	
(2)	$R_{\iota_2(\mathbf{1},\iota_2(\mathbf{1},\iota_1))}xy_1y_2y_3\checkmark\,,\neg\neg p : y_1\checkmark\,,\neg\neg q : y_2\checkmark\,,\neg\neg[\mathbf{1}(\iota_2)](\neg p\,,\neg q) : y_3\checkmark$	$(\langle\alpha\rangle E)$ from (1)
(3)	$p : y_1$	$(\neg E)$ from (2)
(4)	$q : y_2$	$(\neg E)$ from (2)
(5)	$[1(\iota_2)](\neg p\,,\neg q) : y_3\checkmark$	$(\neg E)$ from (2)
(6)	$R_{\iota_2}xz_1z_2\,,R_{\mathbf{1}}z_1y_1\,,R_{\iota_2(\mathbf{1},\iota_1)}z_2y_2y_3\checkmark$	(RR) from (2)
(7)	$R_{\iota_2}z_2w_1w_2\,,R_{\mathbf{1}}w_1y_2\,,R_{\iota_1}w_2y_3$	(RR) from (6)
(8)	$[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : y_3$	$([\alpha(\beta)]D)$ from (5)
(9)	$[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : w_2$	$(R_{\iota_n}x)$ from (7) and (8)
(10)	$[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : z_2$	$(R_{\iota_n}x)$ from (7) and (9)
(11)	$R_{\mathbf{1}}z_2y_2$	$(R_{\iota_n}R_\alpha)$ from (7)
(12)	$[\iota_2](\neg p\,,\neg q) : y_2$	$([\alpha]E)$ from (11) and (10)
(13)	$R_{\iota_2}y_2y_2y_2$	$(R_{\iota_2}I)$
(14)	$\neg q : y_2$ $\quad$ $\neg p : y_2$	$([\alpha]E)$, (14a) closes w.r.t. (4)
(15)	$\otimes$ $\quad$ $[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : x$	$(R_{\iota_n}x)$ from (6) and (10)
(16)	$R_{\mathbf{1}}xy_1$	$(R_{\iota_n}R_\alpha)$ from (6)
(17)	$[\iota_2](\neg p\,,\neg q) : y_1$	$([\alpha]E)$ from (16) and (15)
(18)	$R_{\iota_2}y_1y_1y_1$	$(R_{\iota_2}I)$
(19)	$\neg p : y_1$ $\quad$ $\neg q : y_1$	$([\alpha]E)$, (19a) closes w.r.t. (3)
	$\otimes$	

We continue the tableau from (19).

(19)	$\neg q : y_1$	
(20)	$[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : w_1$	$(R_{\iota_n}x)$ from (7) and (9)
(21)	$[\mathbf{1}]([\iota_2](\neg p\,,\neg q)) : z_1$	$(R_{\iota_n}x)$ from (6) and (10)
(22)	$R_{\mathbf{1}}z_2y_1$	$(R_{\iota_n}R_\alpha)$ from (6)
(23)	$R_{\mathbf{1}}w_1y_1$	$(R_{\iota_n}R_\alpha)$ from (7) and (22)
(24)	$R_{\mathbf{1}}w_2y_1$	$(R_{\iota_n}R_\alpha)$ from (7) and (22)
(25)	$R_{\mathbf{1}}y_3y_1$	$(R_{\iota_n}R_\alpha)$ from (7) and (24)
(26)	$R_{\mathbf{1}}w_2y_2$	$(R_{\iota_n}R_\alpha)$ from (7)
(27)	$R_{\mathbf{1}}y_3y_2$	$(R_{\iota_n}R_\alpha)$ from (7) and (26)
(28)	$R_{\mathbf{1}}z_1y_2$	$(R_{\iota_n}R_\alpha)$ from (6) and (11)
(29)	$R_{\mathbf{1}}xy_2$	$(R_{\iota_n}R_\alpha)$ from (6) and (11)

We will stop the tableau here. This is for presentation reasons. The tableau will reach all its end nodes without closing. To see this, notice that none of $(\neg E), (\langle\alpha\rangle E), (RR), ([\alpha(\beta)]D)$ are applicable to any non-ticked tableau terms occurring in the tableau. Also, for any application of $(R_{\iota_n}x)$, it would repeat what is already on the branch. This is the same with the $([\alpha]E)$ rule. We can still apply the $(R_{\iota_1}I)$ and $(R_{\iota_2}I)$ rules, but this adds no 'useful' information. Now the $(R_{\iota_n}R_\alpha)$ rule can still be applied to the relational atoms that have ι_1 or ι_2 as their underlying relation, but displaying this will result in a tableau that has more than two hundred steps. This is because $x, y_3, z_1, z_2, w_1, w_2$ are related to each other with some ι-relation and so $(R_{\iota_n}R_\alpha)$ will have to be applied multiple times so that every combination of $R_{\iota_2} *{*}*$ and $R_{\iota_1} *{*}$ can be listed — $6 \times 6 \times 6 > 200$. Once this is done, every rule will be used up to saturation. Since this tableau is open, we can find a model to satisfy the original formula. But to say this with confidence, we need completeness.

We reiterate what the $\checkmark$'s indicate with an example. To create node (6), we apply (RR) using $R_{\iota_2(\mathbf{1},\iota_2(\mathbf{1},\iota_1))}xy_1y_2y_3$ from the displayed node (2). This relational atom is replaced with three new relational atoms, and so $R_{\iota_2(\mathbf{1},\iota_2(\mathbf{1},\iota_1))}xy_1y_2y_3$ can be ticked off. What this means is that $R_{\iota_2(\mathbf{1},\iota_2(\mathbf{1},\iota_1))}xy_1y_2y_3$ is not part of the implicit Δ in node (6).

These are all the examples that we will do in this section. We have assumed soundness and completeness of **PMT**, but the next two sections are dedicated to proving the soundness and completeness of **PMT**.

1.4 Soundness of PMT

This section is dedicated to proving the soundness of **PMT**. We need to prove that $\Gamma \vdash_{\mathbf{PMT}} \phi$ implies $\Gamma \Vdash \phi$. So we may assume $\Gamma \vdash_{\mathbf{PMT}} \phi$. I.e., when any saturated tableau with $\{\neg\phi : x\} \cup \{\gamma_i : x \mid \gamma_i \in \Gamma\}$ in the root, is closed. This means that, according to Definition 1.2.8, each branch contains both $\psi : y$ and $\neg\psi : y$ or it contains $\neg[\iota_0] : y$ for some $\psi \in \mathcal{L}_{P(\tau)}, y \in$ Names. Now, to prove soundness, we have to prove that what is in the root, i.e., $\Gamma \cup \{\neg\phi\}$, is also not satisfiable. So conversely, we can prove soundness by proving that if $\Gamma \cup \{\neg\phi\}$ is satisfiable, then at least one of the leaf nodes is satisfiable.

Theorem 1.4.1. ***PMT*** *is sound. I.e.,* $\Gamma \vdash_{\boldsymbol{PMT}} \phi$ *implies* $\Gamma \Vdash \phi$.

Proof. To prove the soundness of **PMT**, we prove that each rule preserves satisfiability from top to bottom. Therefore if we have $\Gamma \vdash_{\mathbf{PMT}} \phi$ and our rules preserve satisfiability, then $\Gamma \Vdash \phi$.

So let's prove the soundness of each rule. Note, Definition 1.1.6 and Definition 1.2.11 will be used in each of these proofs. We let $\mathfrak{M} = (W, \{R_\alpha\}_{MT(\tau)}, V)$ be an arbitrary $MT(\tau)$-model.

Negation Elimination:

$$\begin{array}{ll} \Delta\,, \neg\neg\phi : x & \\ \quad \Delta\,, \phi : x & \qquad (\neg E) \end{array}$$

Assume $\Delta \cup \{\neg\neg\phi : x\}$ is satisfied in $\mathfrak{M}$. We need to show that $\phi : x$ is satisfied in $\mathfrak{M}$. So since $\{\neg\neg\phi : x\}$ is satisfied in $\mathfrak{M}$, then $\mathfrak{M}, \mathsf{S}(x) \Vdash \neg\neg\phi$, then $\mathfrak{M}, \mathsf{S}(x) \nVdash \neg\phi$. Therefore $\mathfrak{M}, \mathsf{S}(x) \Vdash \phi$. Thus $\Delta \cup \{\phi : x\}$ is satisfied in $\mathfrak{M}$.

Diamond Elimination:

$$\begin{array}{ll} \Delta\,, \neg[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x & \\ \Delta\,, R_\alpha x y_1 \ldots y_m\,, \neg\phi_1 : y_1\,, \ldots\,, \neg\phi_m : y_m & \qquad (\langle\alpha\rangle E) \end{array}$$

Where $y_1, \ldots, y_m \in$ Names that are distinct from Names$(\Delta) \cup \{x\}$.

Let $\Delta' = \Delta \cup \{\neg[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x\}$ and $\Delta'' = \Delta \cup \{R_\alpha x y_1 \ldots y_m\,, \neg\phi_1 : y_1\,, \ldots\,, \neg\phi_m : y_m\}$.

Now, assume Δ' is satisfied in $\mathfrak{M}$. This means that $\mathfrak{M}, \mathsf{S}(x) \Vdash \neg[\alpha](\phi_1, \phi_2, \ldots, \phi_m)$. Then $\mathfrak{M}, \mathsf{S}(x) \nVdash [\alpha](\phi_1, \phi_2, \ldots, \phi_m)$. I.e., it is not the case that for all $z_1, \ldots, z_m \in W$ with $R_\alpha^{\mathfrak{M}}(\mathsf{S}(x) z_1 \ldots z_m)$ we have $\mathfrak{M}, z_1 \Vdash \phi_1$ or $\mathfrak{M}, z_2 \Vdash \phi_2$ or $\cdots$ or $\mathfrak{M}, z_m \Vdash \phi_m$. Therefore, we have that there exists $z_1 \ldots z_m \in W$ with $R_\alpha^{\mathfrak{M}}(\mathsf{S}(x) z_1 \ldots z_m)$ such that $\mathfrak{M}, z_1 \nVdash \phi_1$ and $\mathfrak{M}, z_2 \nVdash \phi_2$ and $\cdots$ and $\mathfrak{M}, z_m \nVdash \phi_m$. I.e., there exists $z_1 \ldots z_m \in W$ with $R_\alpha^{\mathfrak{M}}(\mathsf{S}(x) z_1 \ldots z_m)$ such that $\mathfrak{M}, z_1 \Vdash \neg\phi_1$ and $\mathfrak{M}, z_2 \Vdash \neg\phi_2$ and $\cdots$ and $\mathfrak{M}, z_m \Vdash \neg\phi_m$.

Now to prove that $\mathfrak{M}$ satisfies Δ'' we must find mappings S' and R' such that it satisfies Definition 1.2.11. To do this, we let $\mathsf{R}'|_{\mathrm{RNames}(\Delta')} = \mathsf{R}|_{\mathrm{RNames}(\Delta')}$ and set $\mathsf{R}'(R_\alpha) = R_\alpha^{\mathfrak{M}}$. We also let S' be such that: $\mathsf{S}'|_{\mathrm{Names}(\Delta')} = \mathsf{S}|_{\mathrm{Names}(\Delta')}$ and set $\mathsf{S}'(y_i) = z_i$ for $1 \le i \le m$, since all y_i's are new. Using R' and S', we can conclude $\mathfrak{M}$ satisfies Δ'' and hence the rule is sound.

Box Elimination:

$$\begin{array}{c} \Delta\,, [\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x\,, R_\alpha x y_1 \ldots y_m \\ \Delta'\,, \phi_1 : y_1 \quad \cdots \quad \Delta'\,, \phi_m : y_m \qquad ([\alpha]E) \end{array}$$

Where Δ' stands for $\Delta\,, [\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x\,, R_\alpha x y_1 \ldots y_m$

This rule branches, so to prove soundness, we need to show that assuming the premise is satisfied in $\mathfrak{M}$ results in at least one of the conclusions being satisfied in $\mathfrak{M}$.

Assume $\Delta \cup \{[\alpha](\phi_1\,, \phi_2\,, \ldots\,, \phi_m) : x, R_\alpha x y_1 \ldots y_m\}$ is satisfied in $\mathfrak{M}$. Thus $\mathfrak{M}, \mathsf{S}(x) \Vdash [\alpha](\phi_1, \phi_2, \ldots, \phi_m)$, and since we also have $R_\alpha(x y_1 \ldots y_m)$ then, $(\mathsf{S}(x), \mathsf{S}(y_1), \ldots, \mathsf{S}(y_m)) \in R_\alpha^{\mathfrak{M}}$, i.e., $R_\alpha^{\mathfrak{M}}(\mathsf{S}(x)\mathsf{S}(y_1) \ldots \mathsf{S}(y_m))$. Therefore, we have $\mathfrak{M}, \mathsf{S}(y_1) \Vdash \phi_1$ or $\mathfrak{M}, \mathsf{S}(y_2) \Vdash \phi_2$ or $\cdots$ or $\mathfrak{M}, \mathsf{S}(y_m) \Vdash \phi_m$. And so, at least one of these is true, hence the rule is sound.

ι_n-relation on a state:
For some $0 \leq i, j \leq n \in \{1, 2\}, i \neq j$

$$\frac{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, \phi : x_i}{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, \phi : x_i\,, \phi : x_j} \qquad (R_{\iota_n} x)$$

Assume $\Delta \cup \{R_{\iota_n}(x_0 \ldots x_n)\,, \phi : x_i\}$ $(0 \leq i \leq n)$ is satisfied in $\mathfrak{M}$. Then we need to show that $\{\phi : x_j\}$ $(0 \leq j \leq n, i \neq j)$ is satisfied in $\mathfrak{M}$. Now by assumption, we have $\mathfrak{M}, \mathsf{S}(x_i) \Vdash \phi$, we also have $R_{\iota_n}(x_0 \ldots x_n)$ so this means that $(\mathsf{S}(x_0), \ldots, \mathsf{S}(x_n)) \in R^{\mathfrak{M}}_{\iota_n} = \{(w, \ldots, w) \mid w \in W\}$. Therefore $\mathsf{S}(x_0) = \cdots = \mathsf{S}(x_n)$. So, $\mathfrak{M}, \mathsf{S}(x_j) \Vdash \phi$.

ι_n-relation on any other relation:
For some $0 \leq i, j \leq n \in \{1, 2\}, i \neq j$

$$\frac{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, R_\alpha(b_0 b_1 \ldots x_i \ldots b_m)}{\Delta\,, R_{\iota_n}(x_0 \ldots x_n)\,, R_\alpha(b_0 b_1 \ldots x_i \ldots b_m)\,, R_\alpha(b_0 b_1 \ldots x_j \ldots b_m)} \qquad (R_{\iota_n} R_\alpha)$$

Assume $\Delta \cup \{R_{\iota_n}(x_0 \ldots x_n)\,, R_\alpha(b_0 b_1 \ldots x_i \ldots b_m)\}$ $(0 \leq i \leq n)$ is satisfied in $\mathfrak{M}$, then we need to show that $\{R_\alpha(b_0 b_1 \ldots x_j \ldots b_m)\}$ is satisfied in $\mathfrak{M}$. Now, by our assumption, $(\mathsf{S}(b_0), \mathsf{S}(b_1), \ldots, \mathsf{S}(x_i), \ldots, \mathsf{S}(b_m)) \in R^{\mathfrak{M}}_\alpha$, and $(\mathsf{S}(x_0), \ldots, \mathsf{S}(x_n)) \in R^{\mathfrak{M}}_{\iota_n}$. Therefore $\mathsf{S}(x_0) = \cdots = \mathsf{S}(x_n)$. So, $(\mathsf{S}(b_0), \mathsf{S}(b_1), \ldots, \mathsf{S}(x_j), \ldots, \mathsf{S}(b_m)) \in R^{\mathfrak{M}}_\alpha$ and thus $\{R_\alpha(b_0 b_1 \ldots x_j \ldots b_m)\}$ is satisfied in $\mathfrak{M}$.

ι_1-relation introduction:
For some $x \in \text{Names}(\Delta)$

$$\frac{\Delta}{\Delta\,, R_{\iota_1}(xx)} \qquad (R_{\iota_1} I)$$

Assume that Δ is satisfied in $\mathfrak{M}$. We need to show that $\{R_{\iota_1}(xx)\}$ is satisfied in $\mathfrak{M}$. Since we have that $x \in \text{Names}(\Delta)$, then we know that $\mathsf{S}(x) \in W$. Therefore $(\mathsf{S}(x), \mathsf{S}(x)) \in R^{\mathfrak{M}}_{\iota_1}$.

ι_2-relation introduction:
For some $x \in \text{Names}(\Delta)$

$$\frac{\Delta}{\Delta\,, R_{\iota_2}(xxx)} \qquad (R_{\iota_2} I)$$

Assume that Δ is satisfied in $\mathfrak{M}$. We need to show that $\{R_{\iota_2}(xxx)\}$ is satisfied in $\mathfrak{M}$. Since we have that $x \in \text{Names}(\Delta)$, then we know that $\mathsf{S}(x) \in W$. Therefore $(\mathsf{S}(x), \mathsf{S}(x), \mathsf{S}(x)) \in R^{\mathfrak{M}}_{\iota_2}$.

Relation Reduction:

$$\frac{\Delta\,, R_{\alpha(\beta_1, \beta_2, \ldots, \beta_m)}(x x^1_1 x^1_2 \ldots x^1_{\rho(\beta_1)} x^2_1 \ldots x^2_{\rho(\beta_2)} \ldots x^m_1 \ldots x^m_{\rho(\beta_m)})}{\Delta\,, R_\alpha(x y_1 \ldots y_m)\,, R_{\beta_1}(y_1 x^1_1 \ldots x^1_{\rho(\beta_1)})\,, \ldots\,, R_{\beta_m}(y_m x^m_1 \ldots x^m_{\rho(\beta_m)})} \qquad (RR)$$

Where $y_1, \ldots, y_m \in$ Names that are distinct from
$\text{Names}(\Delta) \cup \{x, x^1_1, x^1_2, \ldots, x^1_{\rho(\beta_1)}, x^2_1, \ldots, x^2_{\rho(\beta_2)}, \ldots, x^m_1, \ldots, x^m_{\rho(\beta_m)}\}$
i.e., distinct from any names occurring in the premise.

We will denote $\Delta \cup \{R_{\alpha(\beta_1, \beta_2, \ldots, \beta_m)}(x x^1_1 x^1_2 \ldots x^1_{\rho(\beta_1)} x^2_1 \ldots x^2_{\rho(\beta_2)} \ldots x^m_1 \ldots x^m_{\rho(\beta_m)})\} = \Delta'$ and $\Delta \cup \{R_\alpha(x y_1 \ldots y_m)\,, R_{\beta_1}(y_1 x^1_1 \ldots x^1_{\rho(\beta_1)})\,, \ldots\,, R_{\beta_m}(y_m x^m_1 \ldots x^m_{\rho(\beta_m)})\} = \Delta''$.

Assume Δ' is satisfied in $\mathfrak{M}$, then we have that

$$R^{\mathfrak{M}}_{\alpha(\beta_1, \beta_2, \ldots, \beta_m)}(\mathsf{S}(x)\mathsf{S}(x^1_1)\mathsf{S}(x^1_2) \ldots \mathsf{S}(x^1_{\rho(\beta_1)})\mathsf{S}(x^2_1) \ldots \mathsf{S}(x^2_{\rho(\beta_2)}) \ldots \mathsf{S}(x^m_1) \ldots \mathsf{S}(x^m_{\rho(\beta_m)})).$$

This means that there are states $z_1, z_2, \ldots, z_m \in W$ where, letting $\rho(\beta_i) = b_i$,

$$R^{\mathfrak{M}}_\alpha(\mathsf{S}(x) z_1 \ldots z_m) \text{ and } R^{\mathfrak{M}}_{\beta_i}(z_i \mathsf{S}(x^i_1) \ldots \mathsf{S}(x^i_{b_i}))$$

for all $1 \leq i \leq m$.

Now we claim that Δ'' is satisfied in $\mathfrak{M}$. From the assumption we have that Δ is satisfied in $\mathfrak{M}$, to show that $\{R_\alpha(x y_1 \ldots y_m)\,, R_{\beta_1}(y_1 x^1_1 \ldots x^1_{\rho(\beta_1)})\,, \ldots\,, R_{\beta_m}(y_m x^m_1 \ldots x^m_{\rho(\beta_m)})\}$ is satisfied in $\mathfrak{M}$ we create mappings R' and S'. So, we let $\mathsf{R}' = \mathsf{R}$. We can do this because in Definition 1.2.5 of

RNames if $\alpha(\beta_1,\ldots,\beta_m) \in \mathrm{RNames}(\Delta)$ then $\alpha,\beta_1,\ldots,\beta_m \in \mathrm{RNames}(\Delta)$. We also let S' be such that: $\mathsf{S}'|_{\mathrm{Names}(\Delta')} = \mathsf{S}|_{\mathrm{Names}(\Delta')}$ and set $\mathsf{S}'(y_i) = z_i$ for $1 \leq i \leq m$, since all y_i's are new. Using R' and S', we can conclude $\mathfrak{M}$ satisfies Δ'' and hence the rule is sound.

Boxed Modality Distribution:

$$\frac{\Delta\,,[\alpha(\beta_1\,,\ldots,\beta_{\rho(\alpha)})](\phi_1^1\,,\phi_2^1\,,\ldots,\phi^1_{\rho(\beta_1)}\,,\phi_1^2\,,\ldots,\phi^2_{\rho(\beta_2)}\,,\ldots,\phi_1^m\,,\ldots,\phi^m_{\rho(\beta_m)}):x}{\Delta\,,[\alpha]([\beta_1](\phi_1^1\,,\phi_2^1\,,\ldots,\phi^1_{\rho(\beta_1)})\,,\ldots,[\beta_m](\phi_1^m\,,\ldots,\phi^m_{\rho(\beta_m)})):x}\quad([\alpha(\beta)]D)$$

Assume that $\mathfrak{M}$ satisfies $\Delta \cup \{[\alpha(\beta_1,\ldots,\beta_{\rho(\alpha)})](\phi_1^1,\phi_2^1,\ldots,\phi^1_{\rho(\beta_1)},\phi_1^2,\ldots,\phi^2_{\rho(\beta_2)},\ldots,\phi_1^m,\ldots,\phi^m_{\rho(\beta_m)}):x\}$, therefore,

$$\mathfrak{M},\mathsf{S}(x) \Vdash [\alpha(\beta_1,\ldots,\beta_{\rho(\alpha)})](\phi_1^1,\phi_2^1,\ldots,\phi^1_{\rho(\beta_1)},\phi_1^2,\ldots,\phi^2_{\rho(\beta_2)},\ldots,\phi_1^m,\ldots,\phi^m_{\rho(\beta_m)}).$$

Now, for the sake of contradiction assume that

$$\mathfrak{M},\mathsf{S}(x) \nVdash [\alpha]([\beta_1](\phi_1^1,\phi_2^1,\ldots,\phi^1_{\rho(\beta_1)}),\ldots,[\beta_m](\phi_1^m,\ldots,\phi^m_{\rho(\beta_m)})).$$

This means that there exists states $y_1,\ldots,y_m \in W$ with $R^{\mathfrak{M}}_\alpha(\mathsf{S}(x)y_1\ldots y_m)$ such that $\mathfrak{M},y_i \nVdash [\beta_i](\phi_1^i,\ldots,\phi^i_{\rho(\beta_m)})$, for all $1 \leq i \leq m$. I.e., that for each $1 \leq i \leq m$ we have that $\mathfrak{M},y_i \nVdash [\beta_i](\phi_1^i,\ldots,\phi^i_{\rho(\beta_m)})$. This means that (for each $1 \leq i \leq m$) there exists states $z_1^i,\ldots,z^i_{\rho(\beta_i)} \in W$ with $R^{\mathfrak{M}}_{\beta_i}(y_i z_1^i \ldots z^i_{\rho(\beta_i)})$ such that $\mathfrak{M},z_j^i \nVdash \phi_j^i$, for all $1 \leq j \leq \rho(\beta_i)$.

So this means that we have states $z_1^1,z_2^1,\ldots,z^1_{\rho(\beta_1)},z_1^2,\ldots,z^2_{\rho(\beta_2)},\ldots,z_1^m,\ldots,z^m_{\rho(\beta_m)} \in W$ such that $\mathfrak{M},z_j^i \nVdash \phi_j^i$, for all $1 \leq j \leq \rho(\beta_i)$, $1 \leq i \leq m$. But also $R^{\mathfrak{M}}_\alpha(\mathsf{S}(x)y_1\ldots y_m)$ and that $R^{\mathfrak{M}}_{\beta_i}(y_i z_1^i \ldots z^i_{\rho(\beta_i)})$, for all $1 \leq i \leq m$. This means that

$$R^{\mathfrak{M}}_{\alpha(\beta_1,\ldots,\beta_m)}(\mathsf{S}(x)z_1^1z_2^1\ldots z^1_{\rho(\beta_1)}z_1^2\ldots z^2_{\rho(\beta_2)}\ldots z_1^m\ldots z^m_{\rho(\beta_m)}).$$

And since we know

$$\mathfrak{M},\mathsf{S}(x) \Vdash [\alpha(\beta_1,\ldots,\beta_{\rho(\alpha)})](\phi_1^1,\phi_2^1,\ldots,\phi^1_{\rho(\beta_1)},\phi_1^2,\ldots,\phi^2_{\rho(\beta_2)},\ldots,\phi_1^m,\ldots,\phi^m_{\rho(\beta_m)}),$$

by its semantics we have $\mathfrak{M},z_j^i \Vdash \phi_j^i$, for some $1 \leq j \leq \rho(\beta_i)$, $1 \leq i \leq m$. which contradicts the fact that $\mathfrak{M},z_j^i \nVdash \phi_j^i$, for all $1 \leq j \leq \rho(\beta_i)$, $1 \leq i \leq m$.

Thus $\mathfrak{M},\mathsf{S}(x) \Vdash [\alpha]([\beta_1](\phi_1^1,\phi_2^1,\ldots,\phi^1_{\rho(\beta_1)}),\ldots,[\beta_m](\phi_1^m,\ldots,\phi^m_{\rho(\beta_m)}))$ and hence the rule is sound.

Thus, since every rule is sound, we can conclude that **PMT** is a sound set of tableau rules. □

1.5 Completeness of PMT

In this section, we prove the completeness of **PMT**. Instead of showing that **PMT** is complete by showing $\Gamma \Vdash \phi$ implies $\Gamma \vdash_{\mathbf{PMT}} \phi$, we will be proving an equivalent form of completeness. For this, we need to define what **PMT**-consistency is.

Definition 1.5.1. (PMT-inconsistent) Let Γ be a τ-theory. Then we say that Γ is ***PMT**-inconsistent* if the tableau for Γ closes. And Γ is said to be ***PMT**-consistent* if this is not the case, i.e., it contains an open branch.

An equivalent form of completeness is to prove that every **PMT**-consistent τ-theory is satisfiable. This equivalence is analogous to what was proved in Proposition 0.0.4. To prove that every **PMT**-consistent τ-theory is satisfiable, we need to be able to find the correct model that satisfies a **PMT**-consistent τ-theory. Here, we construct the pre-model of such a model.

Definition 1.5.2. (Pre PMT model) Let Γ be a **PMT**-consistent τ-theory and $\mathbf{b}$ be an open downward saturated open branch on its tableau. Let $\Delta = \mathbf{U}(\mathbf{b})$ then the *pre **PMT** model* $\mathfrak{M}_+ = (W_+, R^{\mathfrak{M}_+}_{\alpha\in MT(\tau)}, V^{\mathfrak{M}_+}(p))$ of Δ is defined as follows:

1. $W_+ = \text{Names}(\Delta)$,

2. $R^{\mathfrak{M}_+}_{\alpha \in MT(\tau)} = \{(x, y_1, \dots, y_m) \in \text{Names}(\Delta)^{m+1} \mid \alpha \text{ is atomic}, \rho(\alpha) = m, R_\alpha x y_1 \dots y_m \in \Delta\}$,

3. $V^{\mathfrak{M}_+}(p) = \{x \in \text{Names}(\Delta) \mid p : x \in \Delta\}$.

While also defining $\mathsf{S} : x \mapsto x$ and $\mathsf{R} : R_\alpha \mapsto R^{\mathfrak{M}_+}_\alpha$. For the relations corresponding to composite terms, they are generated by the appropriate composition of atomic relations.

We call this a pre-model because the interpretation of the ι terms may not be correct. The correct interpretation would be that any states that are related under an ι relation would be the same state in the model. Our pre **PMT** model here will create distinct states for states related under an ι relation. We can construct the correct model by quotienting over W^+. This is done in Definition 1.5.7. For now, we prove that if a τ-theory Γ is **PMT**-consistent then it is satisfiable in the pre **PMT** model, but first we prove two lemmas.

Lemma 1.5.3. *Let $\boldsymbol{b}$ be an open downward saturated branch and let $\Delta = \boldsymbol{U}(\boldsymbol{b})$. Let $\mathfrak{M}_+$ be the pre **PMT** model of Δ. Then, $R_\alpha w v_1 \dots v_n \in \Delta$ if and only if $R^{\mathfrak{M}_+}_\alpha w v_1 \dots v_n$ for all relational atoms where α is atomic.*

Proof. This follows from Definition 1.5.2. □

Lemma 1.5.4. *Let $\boldsymbol{b}$ be an open downward saturated branch and let $\Delta = \boldsymbol{U}(\boldsymbol{b})$. Let $\mathfrak{M}_+$ be the pre **PMT** model of Δ. For all relational atoms (specifically the ones that are composite relations), if $R_\alpha w v_1 \dots v_n \in \Delta$, then $R^{\mathfrak{M}_+}_\alpha w v_1 \dots v_n$.*

Proof. We prove this by induction on α. *Base case:* Proven in Lemma 1.5.3.

Now we assume the claim holds true for R_α ($\rho(\alpha) = m > 0$) and R_{β_i} for all $1 \leq i \leq m$ and let $\rho(\beta_i) = b_i$. We are required to prove the claim holds for $R_{\alpha(\beta_1,\dots,\beta_m)}$. Now if $R_{\alpha(\beta_1,\dots,\beta_m)} w v^1_1 \dots v^1_{b_1} \dots v^m_1 \dots v^m_{b_m} \in \Delta$ this implies (by downward saturation and (RR)) that there are $u_1, \dots, u_m \in \text{Names}(\Delta)$ such that $R_\alpha w u_1 \dots u_m \in \Delta$ and $R_{\beta_i} u_i v^i_1 \dots v^i_{b_i} \in \Delta$ for all $1 \leq i \leq m$. So by our IH, $R^{\mathfrak{M}_+}_\alpha w u_1 \dots u_m$ and $R^{\mathfrak{M}_+}_{\beta_i} u_i v^i_1 \dots v^i_{b_i}$ for all $1 \leq i \leq m$. Hence, (semantically) $R^{\mathfrak{M}_+}_{\alpha(\beta_1,\dots,\beta_m)} w v^1_1 \dots v^1_{b_1} \dots v^m_1 \dots v^m_{b_m}$. □

Theorem 1.5.5. *If a τ-theory Γ is **PMT**-consistent, then it is satisfiable in a pre **PMT** model.*

Proof. Let Γ be a **PMT**-consistent τ-theory. Then, after creating a tableau for Γ (Definition 1.2.8), since it is **PMT**-consistent, then we know there will be at least one open branch **b** — Definition 1.5.1. Now, set $\Delta = \mathbf{U}(\mathbf{b})$, we construct the pre **PMT** model as from Definition 1.5.2. We are required to prove that if $\phi : w \in \Delta$ then $\mathfrak{M}_+, w \Vdash \phi$, and for this we use induction on *com* — Definition 1.1.9.

Base Case: Let $\phi = p \in \text{PVAR}$, then $\phi : w \in \Delta$ iff $w \in V^{\mathfrak{M}_+}(p)$ iff $\mathfrak{M}_+, w \Vdash p$ iff $\mathfrak{M}_+, w \Vdash \phi$.

Induction hypothesis: Assume the claim holds for all formulas with strictly lower complexity (*com*) than that of ϕ.

Case 1: $\phi = \neg\psi$, which we split into three cases.

Case 1a: $\psi = \neg\gamma$. If $\phi : w \in \Delta$ we have $\neg\neg\gamma : w \in \Delta$. Now, by downward saturation and $(\neg E)$, $\gamma : w \in \Delta$. Since $com(\gamma) = com(\neg\neg\phi) < com(\phi)$ we can use the induction hypothesis to get $\mathfrak{M}_+, w \Vdash \gamma$. Therefore $\mathfrak{M}_+, w \Vdash \neg\neg\gamma$ and hence $\mathfrak{M}_+, w \Vdash \phi$.

Case 1b: $\psi = [\alpha](\gamma_1, \dots, \gamma_n)$ and α is atomic. If $\phi : w \in \Delta$ we have $\neg[\alpha](\gamma_1, \dots, \gamma_n) : w \in \Delta$. Now, by downward saturation and $(\langle\alpha\rangle E)$, there are $v_1, \dots, v_n \in \text{Names}(\Delta)$ such that $R_\alpha w v_1 \dots v_n \in \Delta$ and $\neg\gamma_i : v_i \in \Delta$ for all $1 \leq i \leq n$.

Note that $com(\phi) = cep(\alpha) + \Sigma^m_{i=1} com(\gamma_i) > com(\gamma_i)$ for all $1 \leq i \leq m$. So by the induction hypothesis (and Lemma 1.5.3) $\mathfrak{M}_+, v_i \Vdash \neg\gamma_i$ and there are $v_1, \dots, v_n \in W_+$ such that $R^{\mathfrak{M}_+}_\alpha w v_1 \dots v_n$. Therefore by its semantics $\mathfrak{M}_+, w \Vdash \neg[\alpha](\gamma_1, \dots, \gamma_n)$, i.e., $\mathfrak{M}_+, w \Vdash \phi$.

Case 1c: $\psi = [\alpha](\gamma_1, \dots, \gamma_n)$ and $\alpha = \pi(\beta_1, \dots, \beta_{\rho(\pi)})$ is composite (let $\rho(\pi) = m, \rho(\beta_i) = b_i$). If $\phi : w \in \Delta$ then we have $\neg[\alpha](\gamma_1, \dots, \gamma_n) : w \in \Delta$. We can reindex the γ's to get

$$\neg[\pi(\beta_1, \dots, \beta_m)](\gamma^1_1, \dots, \gamma^1_{b_1}, \dots, \gamma^m_1, \dots, \gamma^m_{b_m}) : w \in \Delta.$$

Now, by downward saturation and ($\langle\alpha\rangle E$) there are $v_j^i \in \text{Names}(\Delta)$ such that

$$R_{\pi(\beta_1,\ldots,\beta_m)} w v_1^1 \ldots v_{b_1}^1 \ldots v_1^m \ldots v_{b_m}^m \in \Delta \text{ and } \neg\gamma_j^i : v_j^i \in \Delta$$

for all $1 \leq i \leq m$, $1 \leq j \leq b_i$. Therefore, again by downward saturation and (RR), there are $u_i \in \text{Names}(\Delta)$ such that $R_\pi w u_1 \ldots u_m$, $R_{\beta_i} u_i v_1^i, \ldots, v_{b_i}^i \in \Delta$ for all $1 \leq i \leq m$.

Note that $com(\phi) = 1 + cep(\pi(\beta_1, \ldots, \beta_m)) + \Sigma_{i=1}^m \Sigma_{j=1}^{b_i} com(\gamma_j^i) > com(\neg\gamma_j^i)$ for all $1 \leq i \leq m$, $1 \leq j \leq b_i$. This means that by the induction hypothesis (and Lemma 1.5.4) there are $v_j^i \in W_+$ and $u_i \in W_+$ such that

$$R_\pi^{\mathfrak{M}_+} w u_1 \ldots u_m \text{ and } R_{\beta_i}^{\mathfrak{M}_+} u_i v_1^i, \ldots, v_{b_i}^i \text{ and } \mathfrak{M}_+, v_j^i \Vdash \neg\gamma_j^i$$

for all $1 \leq i \leq m$, $1 \leq j \leq b_i$. This implies by the semantics that $\mathfrak{M}_+, u_i \Vdash \neg[\beta_i](\gamma_1^i, \ldots, \gamma_{b_i}^i)$ for all $1 \leq i \leq m$. Hence,

$$\mathfrak{M}_+, w \Vdash \neg[\pi]([\beta_1](\gamma_1^1, \ldots, \gamma_{b_1}^1), \ldots, [\beta_m](\gamma_1^m, \ldots, \gamma_{b_m}^m)).$$

Therefore, $\mathfrak{M}_+, w \Vdash \neg[\pi(\beta_1, \ldots, \beta_m)](\gamma_1^1, \ldots, \gamma_{b_1}^1, \ldots, \gamma_1^m, \ldots, \gamma_{b_m}^m)$ and so, $\mathfrak{M}_+, w \Vdash \neg[\alpha](\gamma_1, \ldots, \gamma_n)$, i.e., $\mathfrak{M}_+, w \Vdash \phi$.

Case 2: $\phi = [\alpha](\psi_1, \ldots, \psi_n)$. Which we split it into two cases.

Case 2a: Let $\phi = [\alpha](\psi_1, \ldots, \psi_n)$ and α be atomic. If $\phi : w \in \Delta$ we have $[\alpha](\psi_1, \ldots, \psi_n) : w \in \Delta$. Now, by downward saturation and ($[\alpha]E$) for all $v_1, \ldots, v_n \in \text{Names}(\Delta)$ if $R_\alpha w v_1 \ldots v_n \in \Delta$ then $\psi_i : v_i \in \Delta$ for some $1 \leq i \leq n$.

Note that $com(\phi) = cep(\alpha) + \Sigma_{i=1}^n com(\psi_i) > com(\psi_i)$ for all $1 \leq i \leq n$. This means that by the induction hypothesis (and Lemma 1.5.3) for all $v_1, \ldots, v_n \in W_+$ if $R_\alpha^{\mathfrak{M}_+} w v_1 \ldots v_n$ then $\mathfrak{M}_+, v_i \Vdash \psi_i$. Therefore, $\mathfrak{M}_+, w \Vdash [\alpha](\psi_1, \ldots, \psi_n)$, i.e., $\mathfrak{M}_+, w \Vdash \phi$.

Case 2b: Let $\phi = [\alpha](\psi_1, \ldots, \psi_n)$ and $\alpha = \pi(\beta_1, \ldots, \beta_{\rho(\pi)})$ is composite (let $\rho(\pi) = m, \rho(\beta_i) = b_i$). If $\phi : w \in \Delta$ we have that $[\alpha](\psi_1, \ldots, \psi_n) : w \in \Delta$. We can reindex the γ's to get

$$[\pi(\beta_1, \ldots, \beta_m)](\psi_1^1, \ldots, \psi_{b_1}^1, \ldots, \psi_1^m, \ldots, \psi_{b_m}^m) : w \in \Delta.$$

Which implies by downward saturation and ($[\alpha(\beta)]D$) that

$$[\pi]([\beta_1](\psi_1^1 \ldots, \psi_{b_1}^1), \ldots, [\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m)) : w \in \Delta.$$

Note that by Lemma 1.1.10, $com([\pi(\beta_1, \ldots, \beta_m)](\psi_1^1, \ldots, \psi_{b_1}^1, \ldots, \psi_1^m, \ldots, \psi_{b_m}^m)) > com([\pi]([\beta_1](\psi_1^1, \ldots, \psi_{b_1}^1), \ldots, [\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m)))$. So by the induction hypothesis

$$\mathfrak{M}_+, w \Vdash [\pi]([\beta_1](\psi_1^1, \ldots, \psi_{b_1}^1), \ldots, [\beta_m](\psi_1^m, \ldots, \psi_{b_m}^m)).$$

Hence, $\mathfrak{M}_+, w \Vdash [\pi(\beta_1, \ldots, \beta_m)](\psi_1^1, \ldots, \psi_{b_1}^1, \ldots, \psi_1^m, \ldots, \psi_{b_m}^m)$. Therefore, $\mathfrak{M}_+, w \Vdash [\alpha](\psi_1, \ldots, \psi_n)$, i.e., $\mathfrak{M}_+, w \Vdash \phi$.

From this we can say that if Γ is a **PMT**-consistent τ-theory then it is satisfiable in a pre-model. □

This theorem may look like a completeness proof, but the issue is that our R_{ι_n} relations are not interpreted correctly in our pre-model. As the correct interpretation of $R_{\iota_n} x_0 x_1 \ldots x_n$ would make $x_0 = x_1 = \cdots = x_n$ in our model. To make this happen, we have to quotient W_+ with an equivalence relation $\sim$ that depends on any ι-relations that are in the branch.

Definition 1.5.6. Let $\mathbf{b}$ be a downward saturated branch in a tableau and let $\Delta = \mathbf{U}(\mathbf{b})$. Now we define a relation ($\sim$) on Names(Δ) with the following rule:

$$u \sim v \text{ if for some } n \in \{1, 2\}, R_{\iota_n} w_0 \ldots u \ldots v \ldots w_n \in \Delta.$$

We claim this is an equivalence relation on Names(Δ).

Proof. We are required to prove the reflexivity, symmetry and transitivity of this relation. Now let u, v, w be arbitrary elements in Names(Δ).

Case 1: From downward saturation $(R_{\iota_1} I)$ would introduce $R_{\iota_1} uu \in \Delta$ and so by definition $u \sim u$.

Case 2: If $u \sim v$ then $R_{\iota_n} x_0 \dots u \dots v \dots x_n \in \Delta$ for some $n \in \{1, 2\}$. From downward saturation $(R_{\iota_1} I)$ would introduce $R_{\iota_1} uu \in \Delta$. Now with the $(R_{\iota_n} R_\alpha)$ rule applied to $R_{\iota_1} uu \in \Delta$ we get $R_{\iota_1} vu \in \Delta$ and so $v \sim u$.

Case 3: If $u \sim v$ then $R_{\iota_n} x_0 \dots u \dots v \dots x_n \in \Delta$ for some $n \in \{1, 2\}$ and from $v \sim w$ we have that $R_{\iota_m} x'_0 \dots v \dots w \dots x'_m \in \Delta$ for some $\in \{1, 2\}$. From here, by downward saturation, the $(R_{\iota_n} R_\alpha)$ rule would be used to get $R_{\iota_n} x'_0 \dots u \dots w \dots x'_m \in \Delta$ and so $u \sim w$. □

We can now define what we call the *Canonical **PMT** model on an open downward saturated branch*, which gives us the correct interpretation of the ι_n model terms. We could define it directly from our tableau, but for the sake of convention and ease of use, we define it based on our pre **PMT** model.

Definition 1.5.7. (Canonical PMT model) We define the *Canonical **PMT** model* as follows. Let **b** be an open downward saturated branch in a tableau and let $\Delta = \mathbf{U}(\mathbf{b})$. We first construct the pre **PMT** model $\mathfrak{M}_+ = (W_+, R^{\mathfrak{M}_+}_{\alpha \in \tau}, V^{\mathfrak{M}_+}(p))$. We construct $\mathfrak{M}_\sim = (W_\sim, R^{\mathfrak{M}_\sim}_{\alpha \in \tau}, V^{\mathfrak{M}_\sim}(p))$ as follows:

1. $W_\sim = W_+ / \sim$
2. $R^{\mathfrak{M}_\sim}_{\alpha \in \tau} = \{([x]_\sim, [y_1]_\sim, \dots, [y_m]_\sim) \in (W_\sim)^{m+1} \mid \alpha \text{ is atomic}, \rho(\alpha) = m, (x, y_1, \dots, y_m) \in R^{\mathfrak{M}_+}_{\alpha \in \tau}\}$
3. $V^{\mathfrak{M}_\sim}(p) = \{[x]_\sim \in W_\sim \mid x \in V^{\mathfrak{M}_+}(p)\}$

Lemma 1.5.8. *The above definition is well-defined, i.e., equivalence classes are independent of the representatives of the elements of Names(Δ). I.e.,*

1. *if $x \sim y$ then $[x]_\sim \in V^{\mathfrak{M}_\sim}(p)$ iff $[y]_\sim \in V^{\mathfrak{M}_\sim}(p)$ for all $p \in PVAR$,*
2. *and for every $\alpha \in MT(\tau)$ and $[u_0]_\sim, \dots, [u_m]_\sim, [v_0]_\sim, \dots, [v_m]_\sim \in W_\sim$, $m \geq 0$, if $u_0 \sim v_0, \dots, u_m \sim v_m$ then $R^{\mathfrak{M}_\sim}_\alpha [u_0]_\sim \dots [u_m]_\sim$ iff $R^{\mathfrak{M}_\sim}_\alpha [v_0]_\sim \dots [v_m]_\sim$.*

Proof. For property one of the above lemma, assume $x \sim y$. Now, $[x]_\sim \in V^{\mathfrak{M}_\sim}(p)$ would only be the case if there is some $z \in W_+$ such that $z \sim x$ and $z \in V^{\mathfrak{M}_+}$. Therefore, $p : z \in \Delta$. Using transitivity of $\sim$ we have that $z \sim y$, hence $R_{\iota_n} w_0 \dots z \dots y \dots w_n \in \Delta$ (for some $n \in \{1, 2\}$). By downward saturation, the $(R_{\iota_n} x)$ rule would have been applied to $p : z \in \Delta$ to produce $p : y \in \Delta$ and so $y \in V^{\mathfrak{M}_\sim}(p)$. Symmetrically, we can argue for the converse by using the symmetry of $\sim$.

For the second property, we have to use induction on $MT(\tau)$. *Base Case:* Let α be atomic with $\rho(\alpha) = m > 0$ and $u_i \sim v_i$ for all $0 \leq i \leq m$. Now, assume $R^{\mathfrak{M}_\sim}_\alpha [u_0]_\sim [u_1]_\sim \dots [u_m]_\sim$. This means that there are $w_0, \dots, w_m$ such that $R^{\mathfrak{M}_+}_\alpha w_0 \dots w_m$ and $w_0 \sim u_0, \dots, w_m \sim u_m$. By transitivity, we have that $w_0 \sim v_0, \dots, w_m \sim v_m$. So, by Definition 1.5.2, $R_\alpha w_0 w_1 \dots w_m \in \Delta$. Note that for each $0 \leq i \leq m$, we have that

$$R_{\iota_{n_i}} x^i_0 \dots w_i \dots v_i \dots x^i_{n_i} \in \Delta$$

for $n_i \in \{1, 2\}$. Through downward saturation we can apply the $(R_{\iota_n} R_\alpha)$ rule $m + 1$ times on $R_\alpha w_0 \dots w_i \dots w_m \in \Delta$ to get

$$R_\alpha v_0 \dots v_i \dots v_m \in \Delta.$$

Since α is atomic, we get that $R^{\mathfrak{M}_+}_\alpha v_0 \dots v_i \dots v_m$. Therefore $R^{\mathfrak{M}_\sim}_\alpha [v_0]_\sim \dots [v_i]_\sim \dots [v_m]_\sim$. Using symmetry of $\sim$, we can argue for the converse.

Inductive Case: Assume for all α such that $cep(\alpha) < k$ that if $u_0 \sim v_0, \dots, u_m \sim v_m$ and $R^{\mathfrak{M}_\sim}_\alpha [u_0]_\sim \dots [u_m]_\sim$ then $R^{\mathfrak{M}_\sim}_\alpha [v_0]_\sim \dots [v_m]_\sim$ for all α such that $cep(\alpha) < k$.

Let α be composite (with $cep(\alpha) = k$) and $\rho(\alpha) = m > 0$ and $u_j \sim v_j$ for all $0 \leq j \leq m$. And assume that $R^{\mathfrak{M}_\sim}_\alpha u_0 u_1 \dots u_m$. Since α is composite we have $\alpha = \pi(\beta_1, \dots, \beta_{\rho(\pi)})$ for some $\pi, \beta_1, \dots, \beta_{\rho(\pi)} \in MT(\tau)$, letting $\rho(\beta_i) = b_i$. Now, after reindexing (both the u's and v's) we have

$$R^{\mathfrak{M}_\sim}_{\pi(\beta_1, \dots, \beta_{\rho(\pi)})} [u_0]_\sim [u^1_1]_\sim \dots [u^1_{b_1}]_\sim \dots [u^{\rho(\pi)}_1]_\sim \dots [u^{\rho(\pi)}_{b_{\rho(\pi)}}]_\sim$$

and so for some $[w_1]_\sim, \ldots, [w_{\rho(\pi)}]_\sim \in W_\sim$ we have that

$$R_\pi^{\mathfrak{M}_\sim}[u_0]_\sim[w_1]_\sim \ldots [w_{\rho(\pi)}]_\sim \text{ and } R_{\beta_i}^{\mathfrak{M}_\sim}[w_i]_\sim[u_1^i]_\sim \ldots [u_{b_i}^i]_\sim$$

for all $1 \leq i \leq \rho(\pi)$. Now we have $u_j \sim v_j$ for $0 \leq j \leq m$, so to use the induction hypothesis, we need to show that $w_i \sim w_i$ for all $1 \leq i \leq \rho(\pi)$, but this is already true since $\sim$ is reflexive. Therefore, by the induction hypothesis, $R_\pi^{\mathfrak{M}_\sim}[v_0]_\sim[w_1]_\sim \ldots [w_{\rho(\pi)}]_\sim$ and $R_{\beta_i}^{\mathfrak{M}_\sim}[w_i]_\sim[v_1^i]_\sim \ldots [v_{b_i}^i]_\sim$ for all $1 \leq i \leq \rho(\pi)$ and so $R_\alpha^{\mathfrak{M}_\sim}[v_0]_\sim[v_1]_\sim \ldots [v_m]_\sim$. Using symmetry of $\sim$, we can argue for the converse. □

Lemma 1.5.9. *Let $\boldsymbol{b}$ be a downward saturated branch and $\Delta = \boldsymbol{U}(\boldsymbol{b})$. Let $\mathfrak{M}_+$ be the pre-model of Δ and $\mathfrak{M}_\sim$ its quotient model. Then for atomic $\alpha \in MT(\tau)$, $R_\alpha^{\mathfrak{M}_+} w_0 w_1 \ldots w_m$ iff $R_\alpha^{\mathfrak{M}_\sim}[w_0]_\sim[w_1]_\sim \ldots [w_m]_\sim$.*

Proof. The left to right direction is by construction. For the right to left direction, assume that $R_\alpha^{\mathfrak{M}_\sim}[w_0]_\sim[w_1]_\sim \ldots [w_m]_\sim$. This means that there are states $v_0, \ldots, v_m \in W_+$ such that $R_\alpha^{\mathfrak{M}_+} v_0 \ldots v_m$ and $v_i \in [w_i]_\sim$ for all $1 \leq i \leq m$. This means that $v_i \sim w_i$ for all $1 \leq i \leq m$, And so by Definition 1.5.2, $R_\alpha v_0 v_1 \ldots v_m \in \Delta$. Note that for each $0 \leq i \leq m$, we have that

$$R_{\iota_{n_i}} x_0^i \ldots v_i \ldots w_i \ldots x_{n_i}^i \in \Delta$$

for $n_i \in \{1, 2\}$. Through downward saturation we can apply the $(R_{\iota_n} R_\alpha)$ rule $m+1$ times on $R_\alpha v_0 \ldots v_i \ldots v_m \in \Delta$ to get

$$R_\alpha w_0 \ldots w_i \ldots w_m \in \Delta.$$

Since α is atomic, we get that $R_\alpha^{\mathfrak{M}_\sim} w_0 \ldots w_i \ldots w_m$. □

We have to show that any formula that is satisfied in the pre **PMT** model is also satisfied in the canonical **PMT** model. We do this by constructing a p-morphism from the pre **PMT** model to the canonical **PMT** model.

Theorem 1.5.10. *Let $\boldsymbol{b}$ be an open downward saturated branch in a tableau and let $\Delta = \boldsymbol{U}(\boldsymbol{b})$. Then let $\mathfrak{M}_+$ and $\mathfrak{M}_\sim$ be the pre and canonical **PMT** models, respectively. Now we prove that $f : \mathfrak{M}_+ \to \mathfrak{M}_\sim$ such that $f(w) = [w]_\sim$ is a p-morphism from $\mathfrak{M}_+$ to $\mathfrak{M}_\sim$.*

Proof. First, we prove that f is a homomorphism, i.e., for all atomic $\alpha \in MT(\tau)$, $\rho(\alpha) = n$, if $(w_0, \ldots, w_n) \in R_\alpha^{\mathfrak{M}_+}$ then $([w_0]_\sim, \ldots, [w_n]_\sim) \in R_\alpha^{\mathfrak{M}_\sim}$. Notice that this is true by construction and by Lemma 1.1.13, it holds for all $\alpha \in MT(\tau)$.

Next we prove that w and $f(w)$ satisfy the same propositional variables. Again, this is true by construction.

Lastly, we need to prove that f respects the back condition for every atomic $\alpha \in MT(\tau)$. Let $w_0 \in W_+$ and $[w_1]_\sim, \ldots, [w_n]_\sim \in W_\sim$ and $\rho(\alpha) = n$. Let $(f(w_0), [w_1]_\sim, \ldots, [w_n]_\sim) = ([w_0]_\sim, [w_1]_\sim, \ldots, [w_n]_\sim) \in R_\alpha^{\mathfrak{M}_\sim}$. Therefore, by Lemma 1.5.9 we have that $(w, w_1, \ldots, w_n) \in R_\alpha^{\mathfrak{M}_+}$. Trivially, $f(w_i) = [w_i]_\sim$ for all $1 \leq i \leq n$, and so the back condition is satisfied. By Lemma 1.1.13, the back condition holds for all $\alpha \in MT(\tau)$.

This shows that f is a p-morphism and $\mathfrak{M}_\sim$ is a p-morphic image of $\mathfrak{M}_+$. □

Theorem 1.5.11. *The system **PMT** is complete w.r.t. the class of all τ-frames.*

Proof. As mentioned in the beginning of this section, we will be proving an equivalent form of completeness. I.e., every **PMT**-consistent τ-theory is satisfiable in a τ-model.

We already laid the groundwork for this. Let Γ be a **PMT**-consistent τ-theory. Then, after creating a tableau for Γ (Definition 1.2.8) since it is **PMT**-consistent, then we know there will be at least one open branch **b** — Definition 1.5.1. Now, set $\Delta = \mathbf{U}(\mathbf{b})$, we construct the pre **PMT** model $\mathfrak{M}_+$ as from Definition 1.5.2. We have proven in Theorem 1.5.5 that there is a state w in $\mathfrak{M}_+$ such that $\mathfrak{M}_+, w \Vdash \Gamma$. With Theorem 1.5.10, we can transfer this satisfaction to the canonical **PMT**-model. I.e., if f is the p-morphism from $\mathfrak{M}_\sim$ to $\mathfrak{M}_+$ then $\mathfrak{M}_\sim, f(w) \Vdash \Gamma$. Since $\mathfrak{M}_\sim$ is a τ-model, with the proper interpretations of any ι-relations, we have proven our original statement. □

Bibliography

[1] Melvin Fitting. Tableau methods of proof for modal logics. *Notre Dame Journal of Formal Logic*, 13(2):237–247, 1972.

[2] Valentin Goranko and Dimiter Vakarelov. Elementary canonical formulae: extending sahlqvist's theorem. *Annals of Pure and Applied Logic*, 141(1-2):180–217, 2006.

[3] Rajeev Goré. Tableau methods for modal and temporal logics. In *Handbook of tableau methods*, pages 297–396. Springer, 1999.

[4] Guido Governatori. Labelled modal tableaux. In Carlos Areces and Robert Goldblatt, editors, *Advances in Modal Logic 7, papers from "Advances in Modal Logic,"*, pages 87–110. College Publications, 2008.

[5] Marcus Kracht and M Kracht. *Tools and techniques in modal logic*, volume 97. Elsevier Amsterdam, 1999.